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Research paper

Research on drift trajectories of typical marine search and rescue objects based on wind-wave-current coupled experiments

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ABSTRACT

Accurate prediction of marine object drift is essential for effective search-and-rescue operations; however, existing models frequently oversimplify the effects of waves, particularly in terms of swell interactions. This study investigates the coupled influence of wind, waves, and currents on drift dynamics through scaled experiments conducted in a hydrodynamic basin equipped with synchronized environmental control systems. Ultra-wideband technology was used to analyze the drift trajectories of five objects—debris, fishing boat, life raft, vertical dummy, and horizontal dummy—under varying wind directions (−60° to 0°) and wave angles (−30° to 30°). The results indicate that waves substantially affect drift velocities and deflection angles. A comparative analysis with the AP98 Leeway model reveals significant discrepancies (average deviations exceeding 6.0 m) when wave effects are disregarded, especially for fishing boats. The study underscores the importance of integrating wave-driven forces, directional spreading, and object geometry into drift models, offering experimental benchmarks to enhance predictive accuracy in multi-physics marine environments. Key findings suggest that incorporating wave-induced drift velocities and object-specific hydrodynamic parameters can improve trajectory prediction accuracy. This work provides empirical benchmarks for refining search-and-rescue models, emphasizing the critical need to account for multidirectional wave interactions in operational frameworks for marine safety and pollution response.

1. Introduction

The intensification of maritime activities, fueled by industrial growth and scientific exploration, has triggered a rise in marine incident occurrences (Kum and Sahin, 2015), frequently ascribed to intricate environmental conditions or operational error (Cao et al., 2023). Effective emergency response hinges on precise drift trajectory prediction, a key component of search and rescue (SAR) operations that delineates the spatial distribution and temporal evolution of drifting objects (Brushett et al., 2017; Kim et al., 2023). Accurate drift forecasting presents a persistent obstacle for ocean engineers, stemming from the complex interaction dynamics among ambient marine forces—specifically wind, wave, and current patterns—and the diverse hydrodynamic responses exhibited by drifting objects. Current models strive to balance spatial coverage with precision to optimize SAR resource allocation, but this balance is impeded by gaps in understanding the complex multi-factor interactions that govern drift mechanics (Córdova and Flores, 2022; Zhu et al., 2021). Drift phenomena are broadly classified into inland and oceanic drift. Inland drift, which occurs in controlled riverine environments, has been extensively investigated using experimental and numerical methods. For example, Jiang et al. (2021) developed a multivariate regression model for oil spill dispersion in rivers based on flume experiments, while Zeng et al. (2022) utilized deep learning techniques to predict buoy trajectories in inland waterways. By contrast, oceanic drift operates within a far more complex system, where wind-wave-current interactions generate non-linear dynamics across expansive spatial scales. While advancements have been made in simulating container drift (Breivik et al., 2012), tracking icebergs (Yulmetov et al., 2016), and modeling human drift (Ličer et al., 2020), existing studies frequently oversimplify the mechanisms of environmental forcing. Notably, the synergistic impact of winds, waves, and currents is seldom quantified, and the influence of

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swells—persistent wave trains with distinct directional properties—remains underexplored. This knowledge gap undermines the operational reliability of search and rescue (SAR) models, particularly in extreme conditions where swells dominate wave fields (Durgadoo et al., 2021; Mu et al., 2024).

Historically, research on oceanic drift has predominantly focused on wind and current influences, driven by their direct measurability and perceived dominance in governing object motion. Early foundational work by Richardson (1997) established empirical relationships between wind speed and ship drift velocity, which were subsequently refined by Anderson et al. (1998) through theoretical frameworks incorporating leeway coefficients. The Leeway model, widely adopted in search and rescue (SAR) operations, partitions wind-induced drift into downwind and crosswind components based on the geometric properties of the object (Breivik and Allen, 2008; Brushett et al., 2017). For instance, Zhou et al. (2020) investigated the effects of wind-current interactions on vessel yaw in port environments, while Tu et al. (2021) validated AP98 model predictions against observed fishing boat trajectories, emphasizing wind's significant role in lateral displacement. Wind and current-driven drift has also been extensively investigated using Lagrangian particle tracking and hydrodynamic models. Zhu et al. (2021) incorporated stochastic sub-grid currents into ensemble trajectory predictions, thereby enhancing the accuracy of buoy drift simulations. However, these studies frequently treat waves as secondary phenomena, either conflating wave-induced Stokes drift with wind-driven surface currents (Daniel et al., 2003) or entirely neglecting wave directionality. For example, Cho et al. (2012) developed a Monte Carlo-based ensemble model for Korean coastal waters that considered wind and tidal currents but disregarded wave effects, attributing trajectory deflections exclusively to turbulent current variability. Such simplifications continue despite evidence indicating that wave-driven forces—especially for partially submerged objects—can substantially modify drift kinematics (Huang and Law, 2013; Burciu, 2007). The heavy reliance on field observations and operational models has further limited the development of mechanistic understanding. Studies such as Krauß et al. (1989), which correlated buoy drift with wind and current measurements, lack controlled experimental conditions to isolate wave contributions. Similarly, Ličer et al. (2020) simulated drowning victim trajectories using atmospheric and oceanic circulation data but did not account for wave-induced accelerations. This omission is concerning, as waves modulate surface layer dynamics through radiation stresses and breaking-induced turbulence—factors that are well-documented to influence drift dispersion (Durgadoo et al., 2021). Consequently, existing models may misrepresent drift behavior in regions characterized by significant swell activity or complex wave spectra.

While the effects of wind and currents dominate drift literature, waves pose unique challenges due to their spectral complexity and ability to induce non-stationary forces. Experimental studies by Huang et al. (2011) demonstrated that wave height and period linearly correlate with the drift velocity of small floating objects in regular waves, a relationship extended to larger objects by Huang and Law (2013). However, these flume-based experiments idealized wave conditions, omitting directional spreading and multi-modal spectra characteristic of real ocean environments. Field studies, such as Yulmetov et al. (2016) iceberg tracking, have empirically associated wave-driven rotation with trajectory deflection but lack generalized frameworks for predictive modeling. The deflection mechanism exemplifies the intricate interplay between wave forces and object geometry. Jia et al. (2016) attributed aircraft debris deflection to asymmetric wave loading, while Burciu (2007) documented life raft leeway variations due to wave-induced rotational moments. These findings highlight the necessity of quantifying divergence angles—the angular offset between drift velocity and wind vectors—across various object types. For example, Mu et al. (2024) reported divergence angles exceeding 15° for fishing boats, in contrast to sub-8° values for life rafts, indicating geometry-dependent wave coupling. However, existing parameterizations, such as those in the Leeway model, fail to systematically incorporate wave-driven divergence, thereby limiting their applicability to objects with high windage areas (Breivik et al., 2012). Crucially, laboratory and computational fluid dynamics (CFD) studies investigating wave-current-wind interactions remain limited. Most wave-drift experiments, such as those conducted by Huang et al. (2011), focus on isolating wave effects rather than exploring their synergies with concurrent wind/current forcing. Similarly, numerical efforts, including Zhu et al.'s (2022) ARMA-based trajectory models, emphasize current variability over wave stochasticity. Physical experiments encounter significant scalability challenges: flumes are incapable of replicating open-ocean wave climates, while wave basins struggle to superimpose controlled wind and current fields (Huang and Law, 2013). Although CFD approaches show promise, they are constrained by high computational costs associated with resolving multi-phase, multi-scale interactions (Zeng et al., 2022). For instance, Jia et al. (2016) combined Leeway and Monte Carlo methods but neglected wave-induced accelerations in their MH370 debris model—an omission indicated by Durgadoo et al. (2021) for underestimating Indian Ocean dispersion patterns.

Currently, the AP98 Leeway model is widely recognized and applied in marine search and rescue operations (Breivik and Allen, 2008; Coppini et al., 2016; Brushett et al., 2017; Medić et al., 2019; Zhu et al., 2019; Zhu et al., 2025). This model was originally proposed by Allan and Plourde in 1999 and later refined by Allen in 2005 (Allan and Plourde, 1999; Allen, 2005). However, in the current AP98 model, the influence of wave action on the object's drift trajectory is rarely taken into account when predicting its movement trajectory.

Extant scholarship accentuates the demand for an exhaustive examination of floating object trajectories subject to the synergistic action of winds, waves, and currents, motivated by three pivotal research deficiencies. Foremost, inadequate empirical data from purpose-built hydrodynamic testing impedes establishing authoritative calibration standards for drift forecast models. Additionally, extant empirical investigations employing environmental simulation infrastructure remain fundamentally restricted to unidirectional hydrodynamic forcing scenarios, offering insufficient consideration of multidirectional wind and wave forcing and limited analysis of the underlying hydrodynamic mechanisms governing drift and lateral deflection. Lastly, existing numerical and empirical models for predicting drift trajectories have primarily focused on wind and current forcing while neglecting wave-induced contributions, particularly the underlying mechanism through which wave action affects drift behavior; the implications of this omission on trajectory accuracy remain poorly quantified. To resolve these gaps, targeted experimental studies elucidating the synergistic interactions of wind, wave, and current forces on drifting objects are imperative for advancing predictive frameworks in search-and-rescue operations, marine pollution response, and offshore safety.

Integrated experimental measurements and numerical analysis characterize drift behavior resulting from interacting atmospheric and hydrodynamic drivers (wind/waves/currents) in this paper. This methodology enables holistic assessment of spatiotemporal evolution characteristics including trajectory geometry, drift velocity, and directional deviations during drift. We investigate the drift trajectory and velocity of various objects within the range of -60-0 degrees wind direction and -30-30 degrees wave direction. Finally, by comparing the predicted trajectories of the AP98 model with the true trajectories, we identify potential areas for improvement in the AP98 model. The research quantifies wind direction and wave direction effects as important factors of drifting objects' positional evolution and velocity characteristics. Wave-induced drift—particularly from swell—is quantitatively substantiated herein, providing validated benchmark essential for enhancing prognostic model considering wave physics.

2. Implementation

2.1. Basin implementation and model geometry

Empirical drift testing occurred within a versatile wind-wave-current basin (60m × 36m × 1.5m operational envelope) at Ocean University of China's Provincial Key Laboratory of Ocean Engineering, as shown in Figs. 1 and 2. A computer-controlled articulated pusher plate wave generator is situated at one end of the basin to simulate target wave conditions. Wave reflection suppression is achieved through energy-

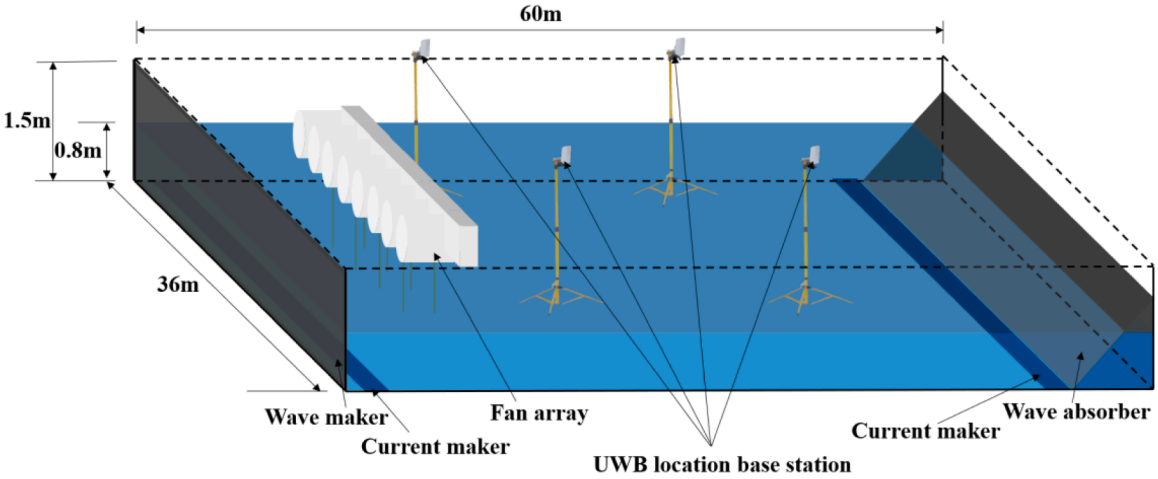

Fig. 1. A layout diagram of the wind-wave-current basin.

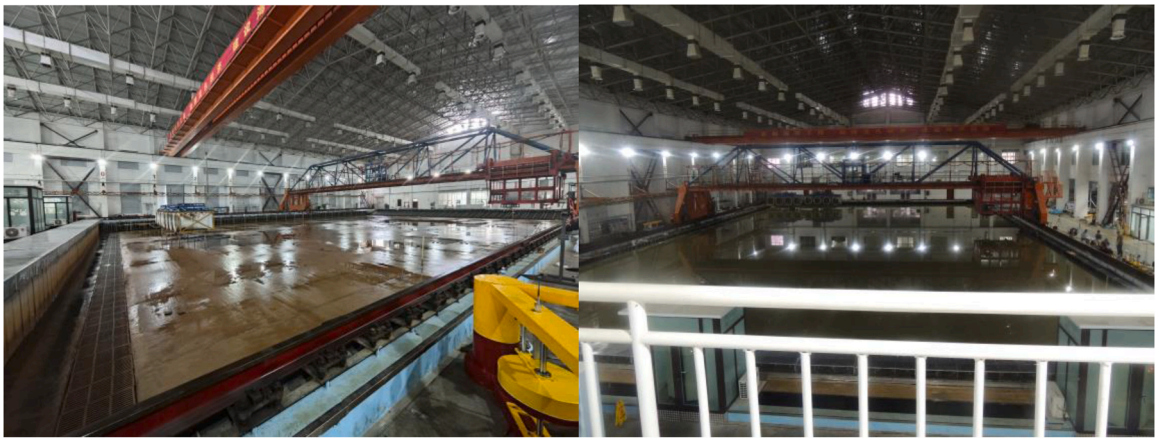

Fig. 2. The global perspective of the wind-wave-current basin.

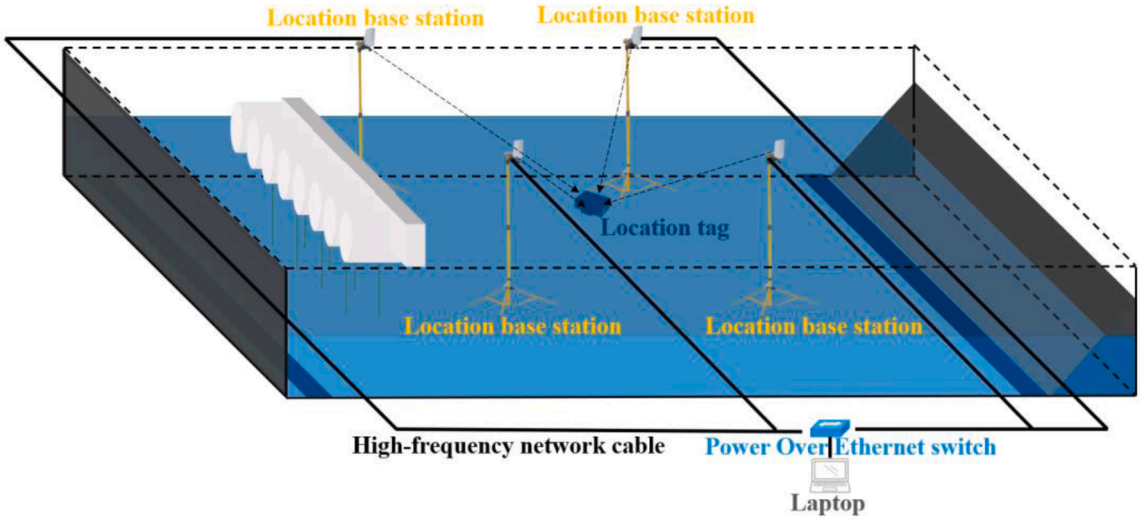

Fig. 3. A layout diagram of the UWB trajectory monitoring system.

dissipative cladding applied to the basin perimeter and terminal boundary. Proximate to the wavemaker, a reversible current pump deploys currents reaching 0.14 m/s at standardized 0.8-meter depth conditions. Secondary current exchange infrastructure interfaces spatially with the terminal absorption zone within the basin's longitudinal extremity. Wind fields are generated by a 15m-spanning modular fan cluster axially aligned with the wavemaker, with motor power controlling wind speed. Preliminary tests determined fan performance envelopes while optimizing UWB (Ultra-Wideband) reference transceiver deployment for high-fidelity drift monitoring.

For trajectory monitoring, a UWB-based system was implemented to capture real-time drift. The setup includes four UWB base stations, six location tags, a laptop, and a video camera, as shown in Fig. 3. The base stations were positioned at the vertices of a predefined experimental zone within the basin, carefully selected to minimize boundary interference from walls and hydrodynamic attenuation effects. Each base station was mounted on height-adjustable tripods (ranging from 0.8 to 3 meters) and secured with counterweights to prevent displacement caused by environmental forces. Signal optimization involved vertically aligning the base station antennas relative to the water surface. Empirical testing determined that an optimal antenna elevation of 1 meter above the waterline provided the best balance between signal integrity and minimal environmental interference.

UWB components were adapted for the challenging air-water interface and hydrodynamic conditions. The base stations and tags were structurally reinforced and waterproofed using polyethylene wraps and specialized tape to protect against splashing and humidity. Network connectivity was ensured through shielded high-frequency Ethernet cables connected to a Power-over-Ethernet (PoE) switch, enabling stable data transmission to the control laptop. After calibration, the system achieved a positioning error of less than 0.1 m and a tag update rate of 10 Hz, satisfying the precision requirements for real-time tracking.

The environmental systems within the basin were synchronized to replicate combined wind-wave-current scenarios. The UWB system tracked tag trajectories across the designated zone, with data processed algorithmically to minimize human intervention. To mitigate signal instability at the air-water interface, antenna orientations were tuned to account for wave surface reflections and wave-induced perturbations.

Key innovations in this setup involve the pioneering application of UWB technology for basin-scale trajectory monitoring and the integration of environmental control systems with real-time positioning infrastructure. This configuration enhances laboratory-scale drift studies by enabling high-resolution, dynamic trajectory analysis under complex multi-physics conditions.

Fig. 4 and Table 1 illustrate the characteristics of the rescue object model. The study utilized scaled physical models to examine the drift patterns of maritime objects under controlled basin conditions. Dimensional scaling ratios of 1:10 and 1:30 were chosen based on Froude and Strouhal similarity criterions to maintain the gravitational effect of free surface current and periodic motion characteristics. The Froude similarity criterion has been robustly validated for modeling combined wind, wave, and current effects, as shown in Fig. 5. For instance, Froude

Table 1
The object model defining characteristics.

Object model	Debris	Fishing boat	Life raft	Vertical dummy	Horizontal dummy
Length (m)	3.47×10^{-1}	8.76×10^{-1}	2.60×10^{-1}	6.10×10^{-2}	1.90×10^{-1}
Beam (m)	1.55×10^{-1}	1.97×10^{-1}	2.60×10^{-1}	3.00×10^{-2}	6.10×10^{-2}
Depth (m)	4.50×10^{-2}	2.96×10^{-1}	1.60×10^{-1}	1.90×10^{-1}	3.00×10^{-2}
Draft (m)	5.00×10^{-3}	1.20×10^{-1}	1.10×10^{-2}	1.50×10^{-1}	2.40×10^{-2}
Displacement (kg)	3.00×10^{-1}	6.74×10^{0}	6.15×10^{-1}	7.00×10^{-2}	7.00×10^{-2}
Moment of inertia I_{xx} ($kg \cdot m^2$)	6.54×10^{-4}	4.42×10^{-2}	4.33×10^{-3}	3.09×10^{-4}	3.90×10^{-5}
Moment of inertia I_{yy} ($kg \cdot m^2$)	8.04×10^{-5}	3.82×10^{-1}	4.33×10^{-3}	5.32×10^{-4}	3.09×10^{-4}
Moment of inertia I_{zz} ($kg \cdot m^2$)	6.34×10^{-4}	3.81×10^{-1}	6.93×10^{-3}	3.90×10^{-5}	5.32×10^{-4}

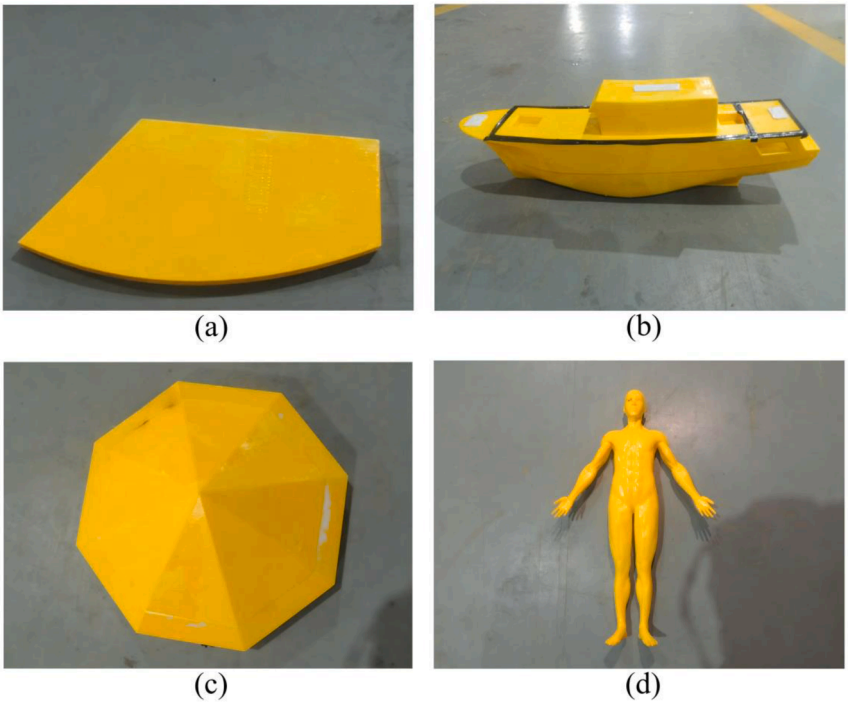

Fig. 4. The object model: (a) debris; (b) fishing boat; (c) life raft; (d) dummy.

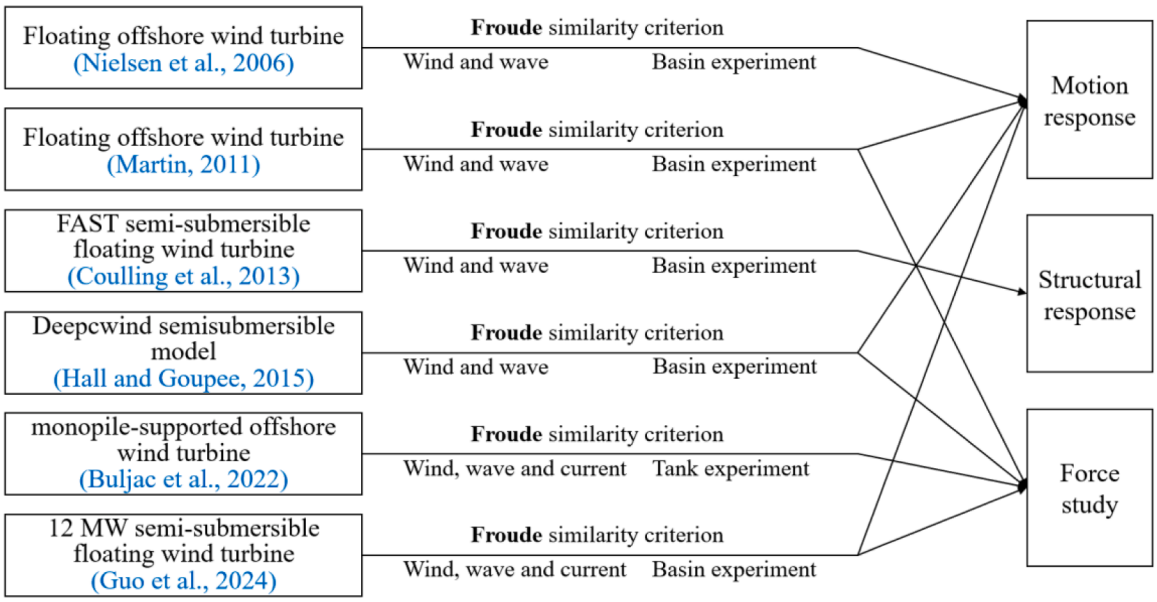

Fig. 5. Reference details.

scaling were successfully applied to monopile turbine (Buljac et al., 2022) and large semi-submersible turbine (Guo et al., 2024) in multi-physics environments. Earlier, its applicability was confirmed to floating turbine dynamics under coupled wind-wave loading (Nielsen et al., 2006; Martin, 2011), while its accuracy was demonstrated in predicting semi-submersible structural responses (Coulling et al., 2013), motion and mooring system forces (Hall and Goupee, 2015). This extensive validation across hydrodynamic, aerodynamic, and structural

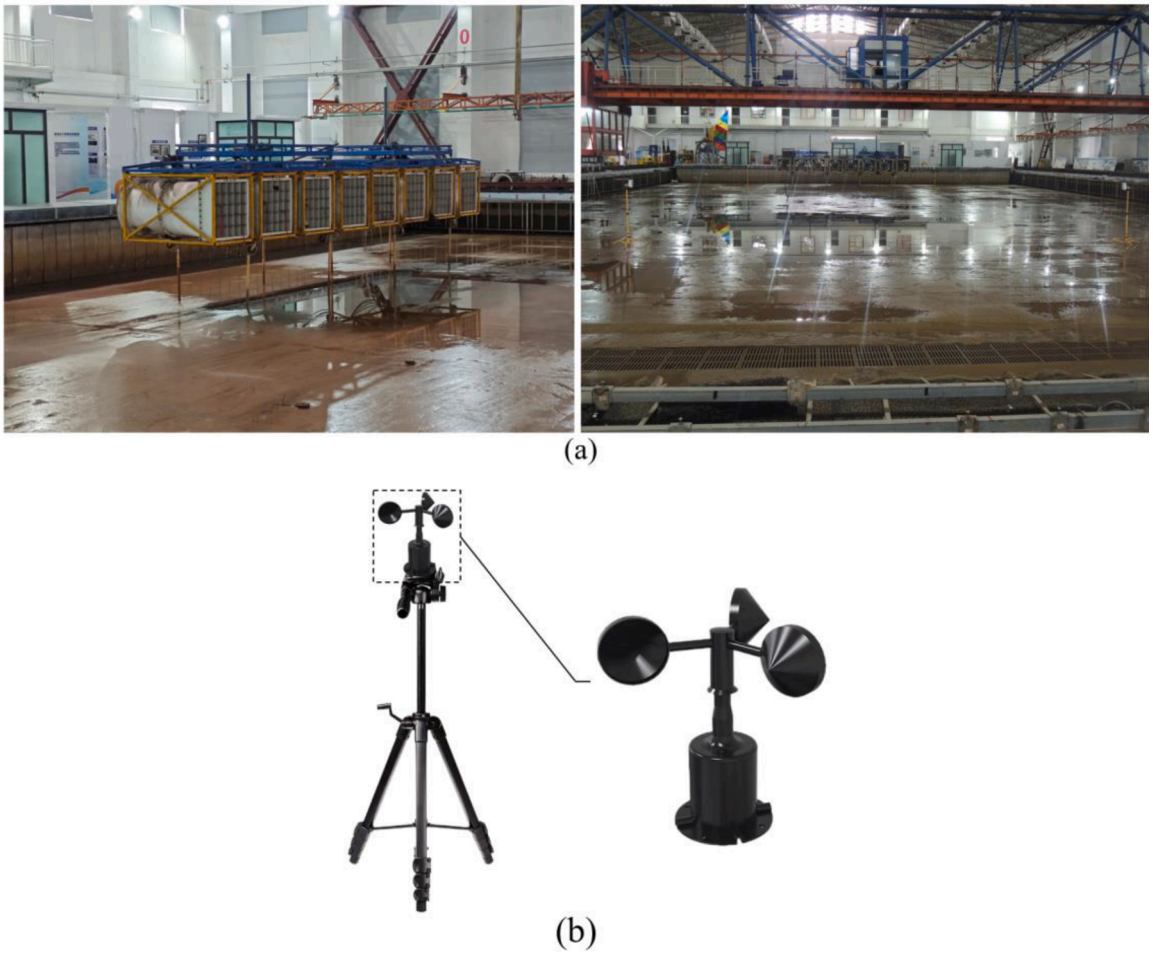

Fig. 6. Fan array and anemometer: (a) fan array; (b) anemometer.

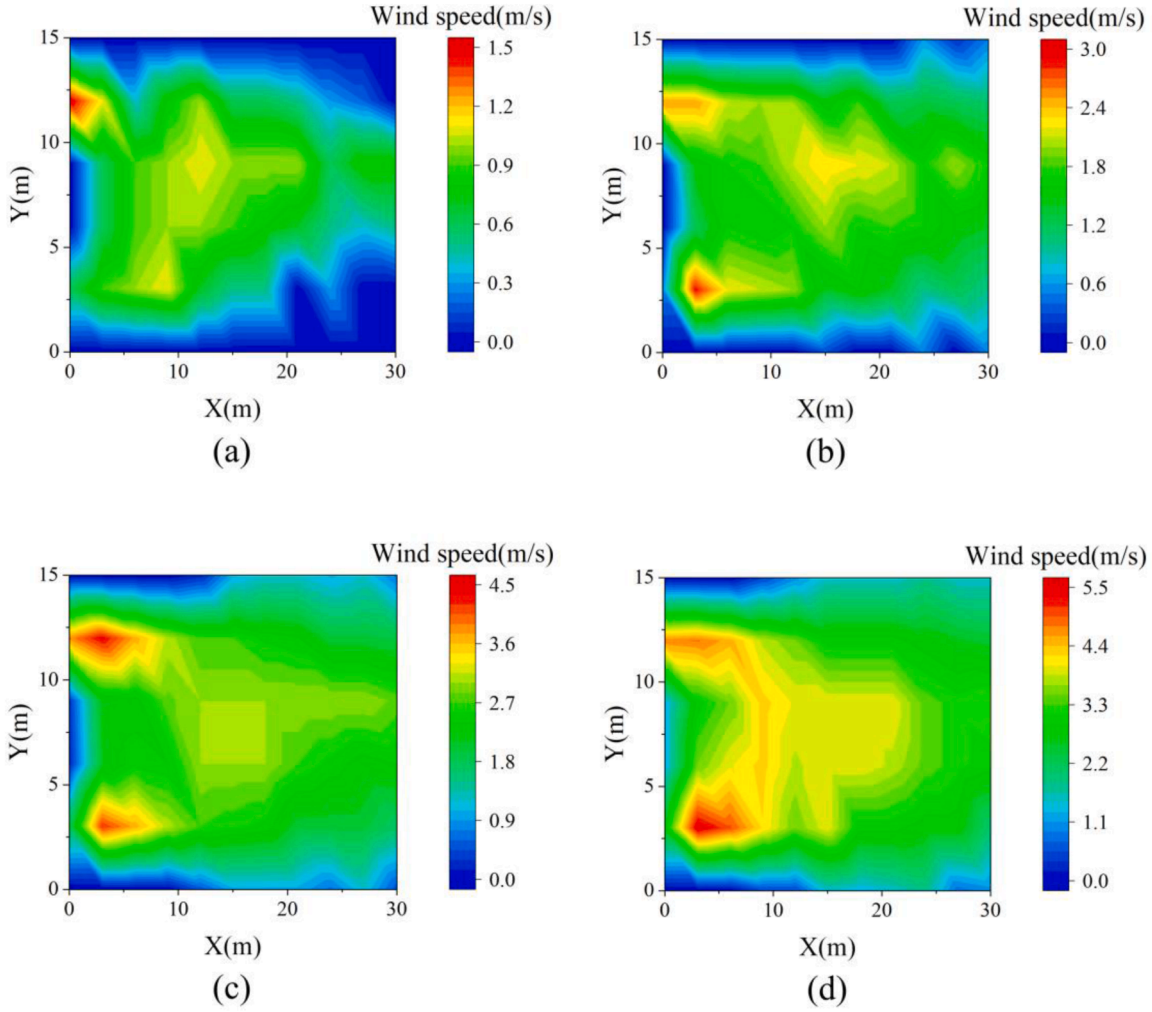

Fig. 7. Measured wind speed of the experimental wind field: (a) 1 m/s reference; (b) 2 m/s reference; (c) 3 m/s reference; (d) 4 m/s reference.

response domains confirms the reliability of the criterion. Furthermore, the cyclical nature of wave-induced object motion necessitates compliance with the Strouhal similarity law. The scale selection was determined by practical limitations of the test facility, including basin dimensions, maximum achievable wind and current speeds, and wave generation capacity: scales of 1:10 were used for debris, life raft, vertical dummy (simulating a life-jacketed human), and horizontal dummy (simulating a surface-floating human), while dimensional proportionality adhered to a 1:30 scaling convention for the fishing boat.

Object models, including debris, fishing boat, life rafts, vertical dummy and horizontal dummy, were fabricated from low-density resin (1.1 g/cm^3) with strategically hollowed structures (3–5 mm wall thickness) to optimize mass distribution. Structural reinforcements were applied at stress-critical zones, such as thickened bilge sections for vessel models and weighted bases for life raft models. A modular design facilitated parameter adjustments, incorporating open-hatch configurations with multi-tiered internal grids for counterweight placement. Symmetrical deck grooves and split-body raft designs ensured stable UWB tag positioning while maintaining watertight integrity through sealed joints of the different parts.

Mass properties were precisely controlled through the implementation of customized counterweight. Comparative testing revealed that water-sand mixtures were optimal for vessel stabilization, as they exhibited non-shifting behavior and allowed for adjustable mass distribution. Human simulators incorporated copper wire weighting in their extremities to replicate realistic buoyancy characteristics; horizontal dummies emphasized uniform mass distribution, whereas vertical dummies concentrated weight in submerged sections. Waterproof double-sided tape and waterproof tape ensured the equipment sealing throughout hydrodynamic testing. All models were rigorously verified for mass, center-of-gravity, and inertia against theoretical values, with 60 g UWB tags incorporated as mass components.

2.2. Environment implementation

The wind generation utilized an array of fans (Fig. 6 (a)). A 0.2m vertical clearance was maintained between the fan array base and the water surface to prevent direct wave interference. Precise wind speed regulation was accomplished through variable motor control, and wind speed measurements were taken 0.1m above the water level using tripod-mounted anemometers to minimize wave disturbance effects (Fig. 6 (b)). This calibration methodology ensured the accurate documentation of wind parameters influencing floating objects

Spatial analysis of the generated wind field (Fig. 7) identified three distinct lateral operational zones:

1. Acceleration zone (0–6 m): Experimental wind velocity datasets register sub-prediction values across validation trials.
2. Stabilization zone (6–21 m): Consistent agreement between measured and theoretical wind speeds.
3. Decay zone (21–30 m): The wind speed is decreasing.

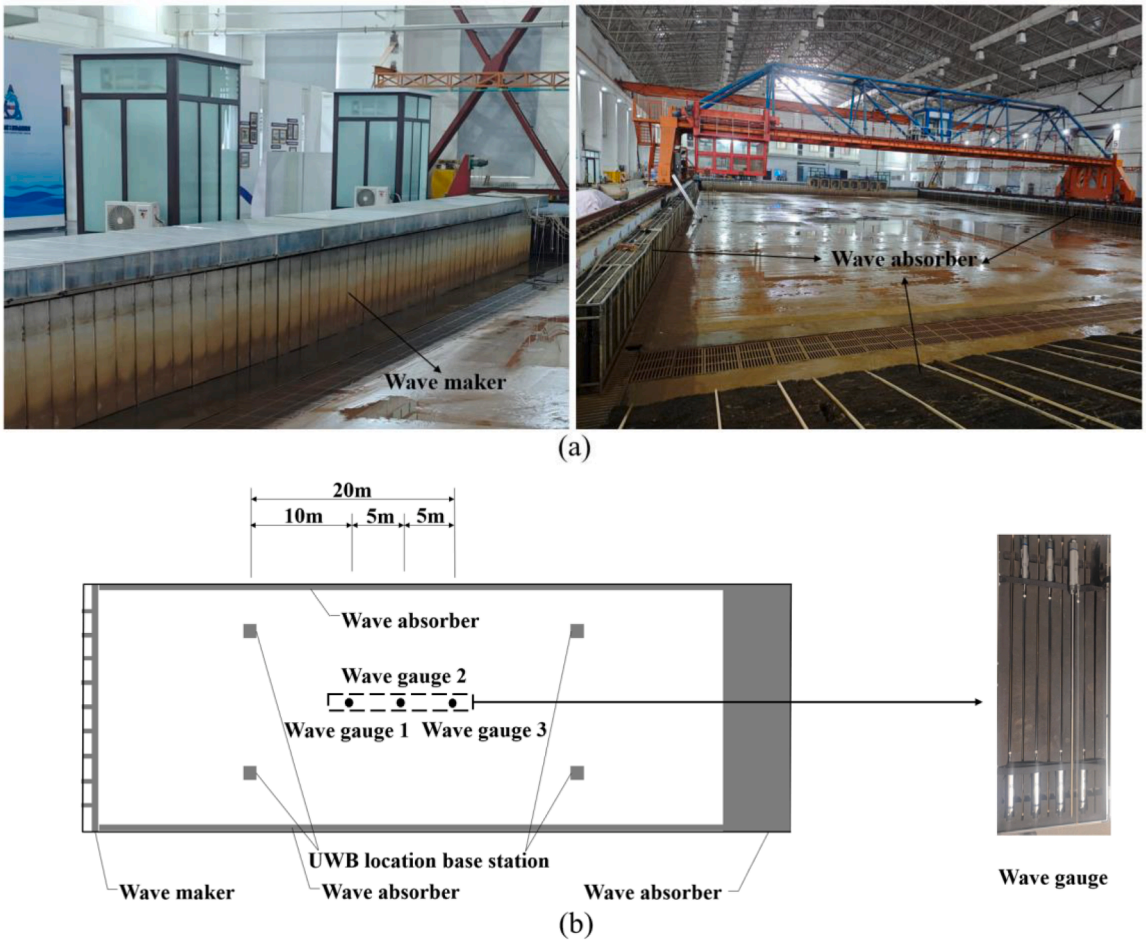

Fig. 8. Wave maker, wave absorber and wave gauge: (a) wave maker and wave absorber; (b) wave gauge.

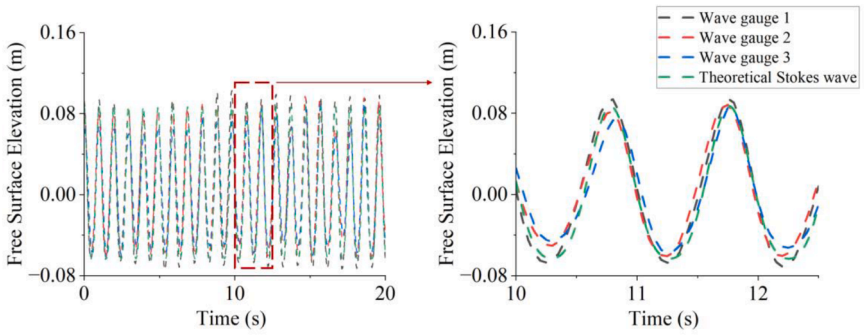

Fig. 9. Empirical versus theoretical waves validation.

Longitudinal measurements indicated maximum wind intensity between -6 m and -9 m offsets from the centerline, with symmetrical attenuation toward the field boundaries. To minimize edge effects, the initial motion position of the object is deployed at the (0, -7.5 m) coordinates. Trajectory analysis confirmed that most of the drift trajectories remained within the stable wind field, while deviations towards the boundary indicated that wind speed gradients had a significant impact on the dynamics of the objects.

The experimental setup featured a computer-controlled pusher plate wave maker integrated with an energy-dissipating beach system (as shown in Fig. 8 (a)). Three wave gauges were strategically placed at the fore, mid, and aft positions along the test section (Fig. 8 (b)) to monitor surface elevation dynamics. For drift analysis, a 3 × 3 parametric matrix of wave characteristics was employed: periods at 0.98, 1.14, 1.29 s intersecting heights of 0.05, 0.10, 0.15 m. Real-time comparisons between the sensor-recorded wave and theoretical Stokes wave predictions facilitated continuous fine-tuning of the wave paddle controller, thereby ensuring waveform accuracy.

Validation trials for the 0.98 s/0.15 m condition revealed the following: (1) Height discrepancies of 8.49 %, 7.45 %, and 16.25 % were observed across measurement stations, with deviations in the core drift zone remaining under 10 % and full-range variations below 20 %. Phase-resolved comparisons (Fig. 9) demonstrated good agreement between experimental and theoretical surface elevations. (2) Progressive wave decay was evident as height reductions of 14.69 % and 22.81 % at subsequent gauges relative to the fore sensor, underscoring significant basin attenuation effects that need to be compensated for in drift analyses.

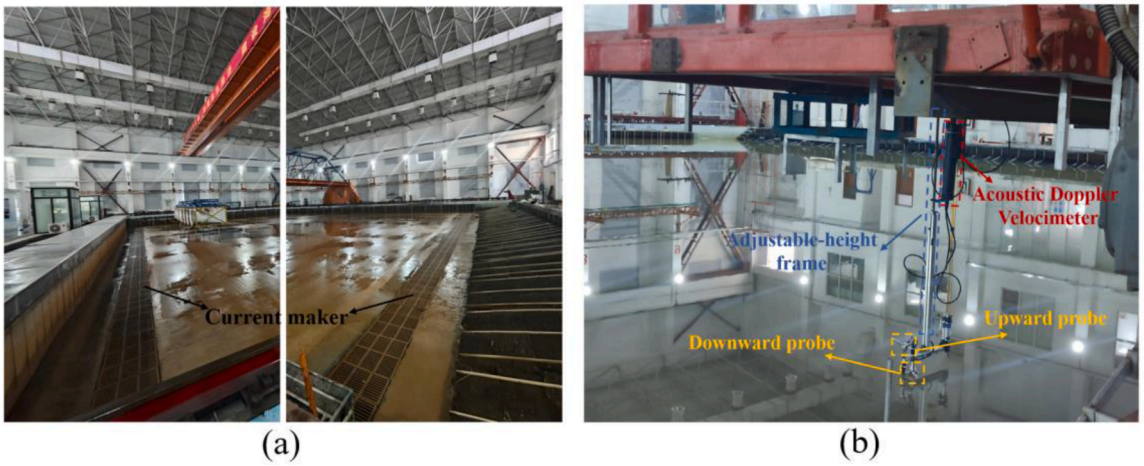

Fig. 10. Current maker and ADV: (a) current maker; (b) ADV.

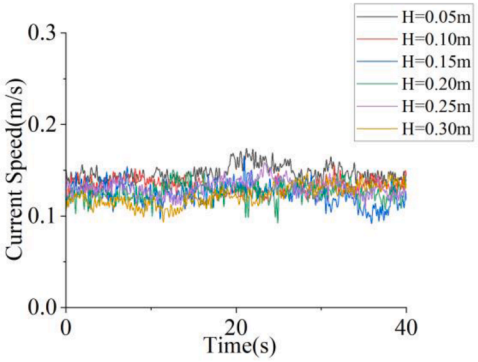

Fig. 11. Empirical current speed.

This calibration protocol focuses not only on waveform recording but also on the rigorous validation of hydrodynamic similarity between experimental conditions and numerical models. The methodology ensures that wave generation errors are maintained below critical thresholds, thereby facilitating subsequent rescue object models drift studies. Special emphasis was placed on spatial decay patterns, which provide crucial input for the application of scaling laws in physical modeling.

The hydrodynamic system utilizes bidirectional current generators (Fig. 10 (a)) to simulate controlled current movements. For drift studies, the calibration of surface currents was conducted using an Acoustic Doppler Velocimeter (ADV) positioned basin's geometric centroid via a trailer-mounted adjustable frame (Fig. 10 (b)). To minimize boundary effects, measurements were restricted to the mid-section of the drift zone. A dual-probe ADV assembly was developed, consisting of upward- and downward-oriented sensors mounted on a height-adjustable support structure. This apparatus, combined with a vibration-damping base, improved measurement precision by reducing mechanical disturbances caused by water turbulence. Automated electronic controls achieved vertical positioning accuracy of ±0.1 mm, allowing for multi-depth velocity profiling.

Three object surface velocities (0.06 m/s, 0.10 m/s, and 0.14 m/s) were selected based on the dimensions of the drift objects. Real-time ADV data integrated into a feedback loop enabled dynamic power adjustments to maintain the preset flow conditions. Calibration protocols prioritized two stability criteria: (1) equilibration of the ADV system with the ambient flow following positioning, and (2) hydrodynamic stabilization after velocity modifications.

High-resolution measurements taken near the air-water interface (0–0.2 m depth) showed close alignment with object velocities (Fig. 11), thereby confirming the experimental validity. The calibration process not only quantified flow parameters but also verified the congruence between empirical observations and hydrodynamic theory. Prioritizing surface currents ensured relevance to drift mechanics.

3. Results and discussions

3.1. Influence of wave on object drift

Wave-induced drift dynamics are dissected across six distinct cases: isolated wind/current fields, wind-wave/wave-current/wind-current combinations, and wind-wave-current coupling. These cases are grouped into three categories based on the presence or absence of waves. In this section, the wind, wave, and current parameters are held constant, and only six combinations are examined. The specific parameter values for wind, waves, and currents are listed as follows: 4 m/s wind speed, 0° wind direction, 0.14 m/s current speed, 0.15 m wave height, 0.98 s wave period, 0° wave incidence angle. 0° azimuth denotes the principal hydrodynamic axis spanning wave maker to terminal wave absorber. Within this convention, clockwise vectors acquire negative magnitude, while counterclockwise increments register positive orientation. Fig. 12 shows the drift experiment process of various objects. Figs. 13, 14, and 15 respectively demonstrate the wave effects on the drift trajectory, x-direction drift velocity, and y-direction drift velocity of debris. Comparative analysis indicates that, under the influence of waves, the drift trajectory is significantly altered, as depicted in Fig. 13. This alteration is primarily due to changes in the y-direction drift velocity caused by wave action, as shown in Fig. 15. Furthermore, the x-direction drift velocity is higher in scenarios with waves compared to those without waves, as demonstrated in Fig. 14.

Figs. 16, 17, and 18 show the wave effects on drift trajectory, x-direction drift velocity, and y-direction drift velocity of fishing boat, respectively. Fig. 16 demonstrates that under wave influence, the drift trajectory undergoes significant alterations. These alterations are primarily attributed to changes in the y-direction drift velocity, as shown in Fig. 18. Furthermore, Fig. 17 indicates that the x-direction drift velocity is greater in case with waves than in those without waves.

Figs. 19, 20, and 21 respectively demonstrate the wave effects on the drift trajectory, x-direction drift velocity, and y-direction drift velocity of the life raft. As shown in Figs. 19 and 21, under wave influence, there are no significant changes in either the drift trajectory or the y-direction drift velocity of the life raft. This phenomenon can be attributed to the life raft's symmetrical shape, which distinguishes it from other rescue objects. Observations from the experimental video reveal that the life raft tends to rotate rather than deviate from its trajectory when drifting in an uneven wind field. Consequently, the impact of waves on the life

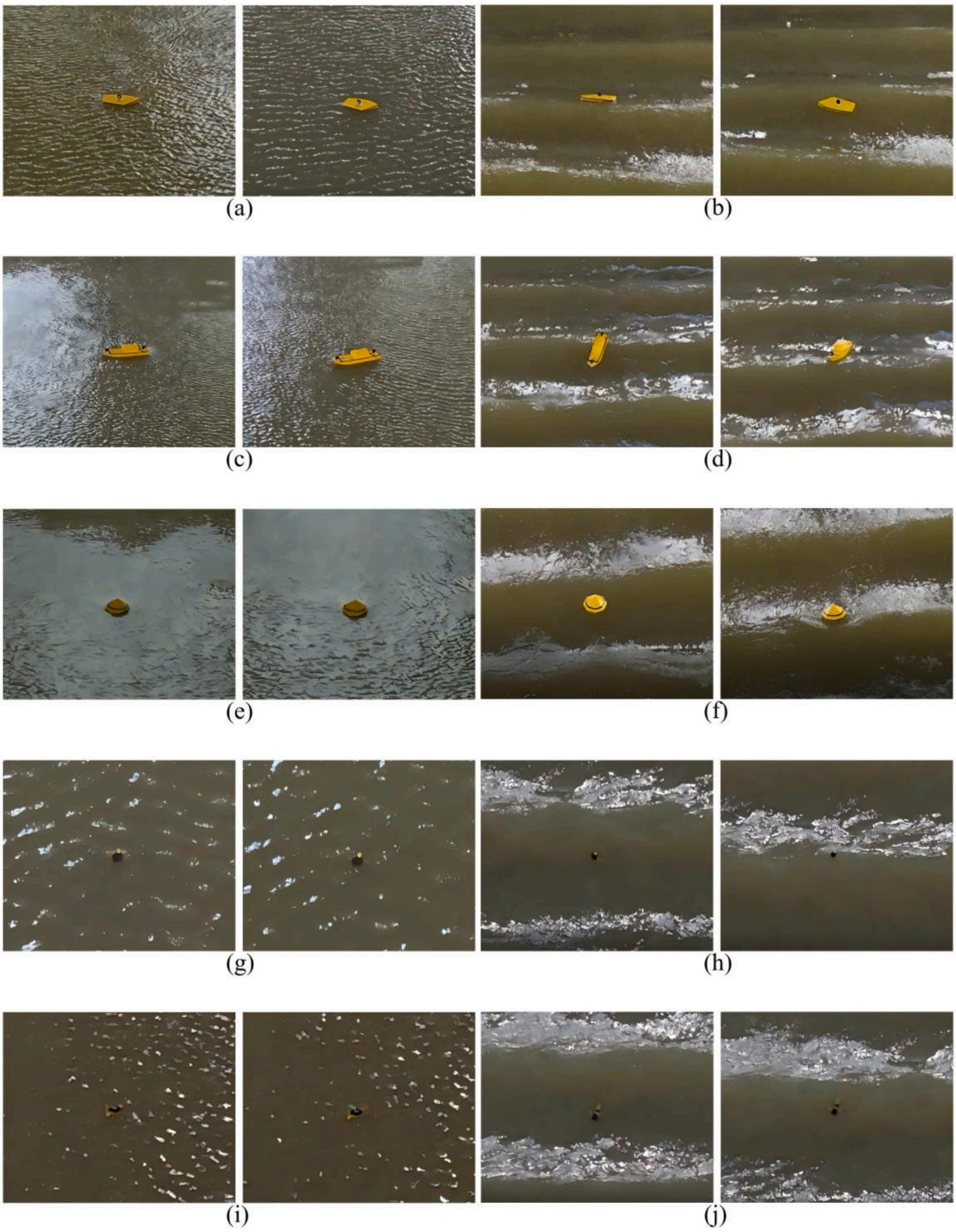

Fig. 12. The rescue object models in the drift experiment (The picture on the left ((a), (c), (e), (g), (i)) is the wind-current case; the picture on the left ((b), (d), (f), (h), (j)) is the wind-wave-current case): (a), (b) debris; (c), (d) fishing boat; (e), (f) life raft; (g), (h) vertical dummy; (i), (j) horizontal dummy.

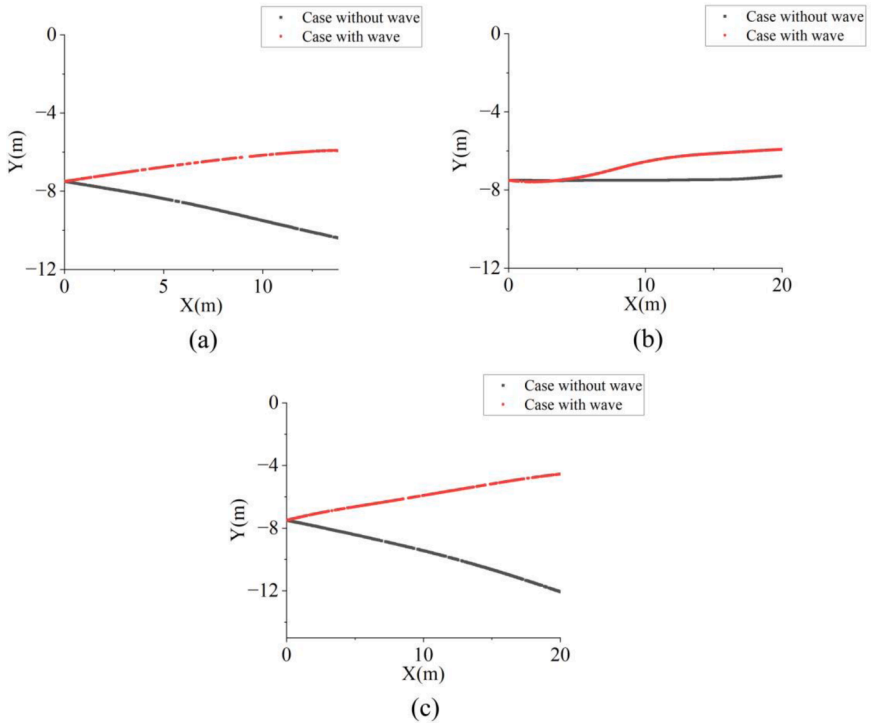

Fig. 13. The wave effects on drift trajectory of the debris: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

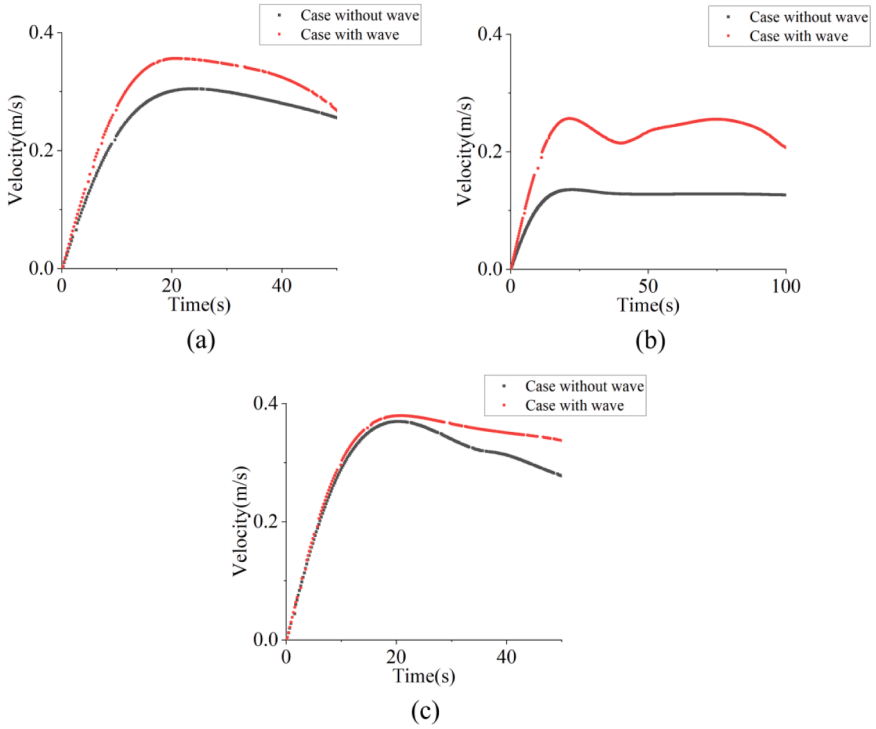

Fig. 14. The wave effects on drift velocity of the debris in the x-direction: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

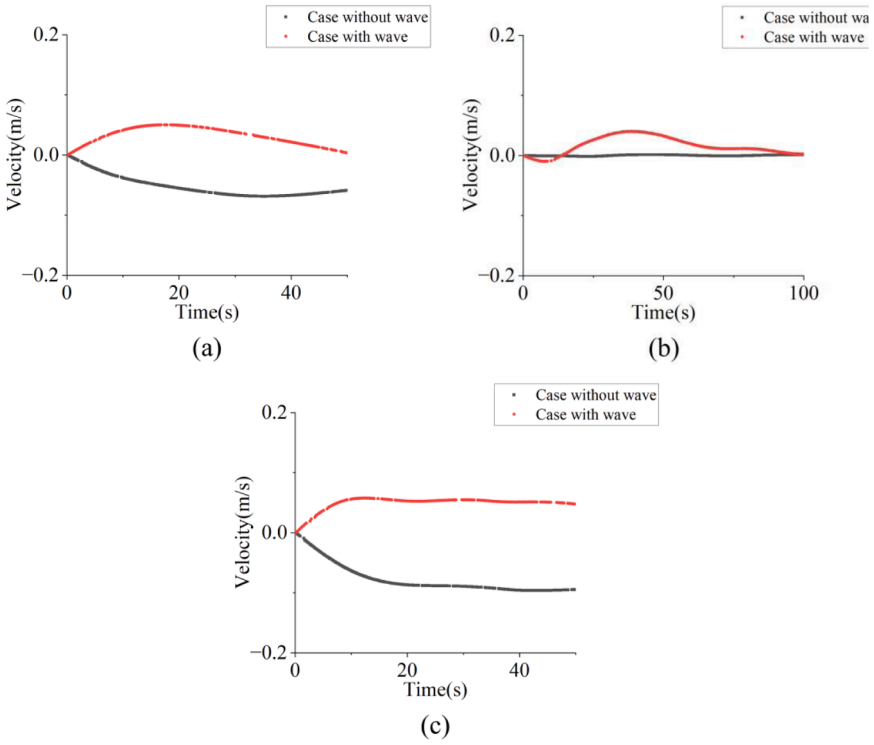

Fig. 15. The wave effects on drift velocity of the debris in the y-direction: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

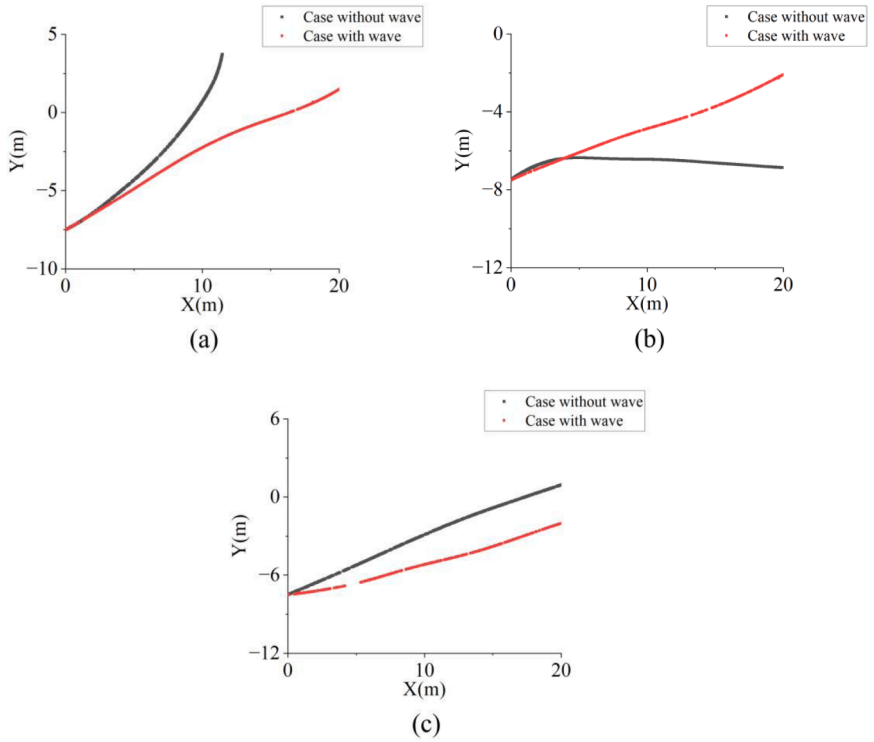

Fig. 16. The wave effects on drift trajectory of the fishing boat: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

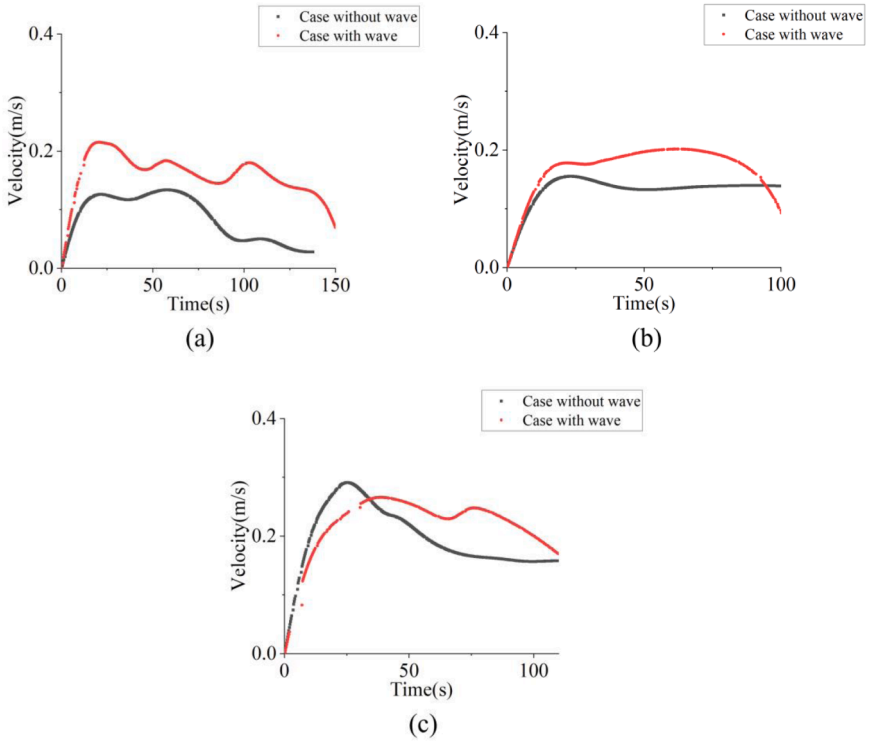

Fig. 17. The wave effects on drift velocity of the fishing boat in the x-direction: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

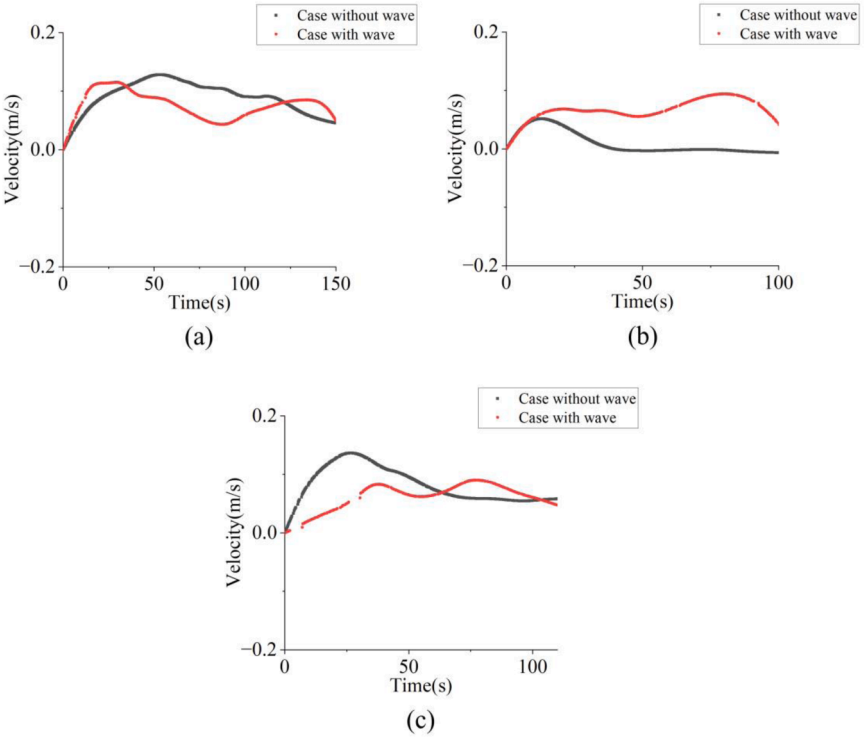

Fig. 18. The wave effects on drift velocity of the fishing boat in the y-direction: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

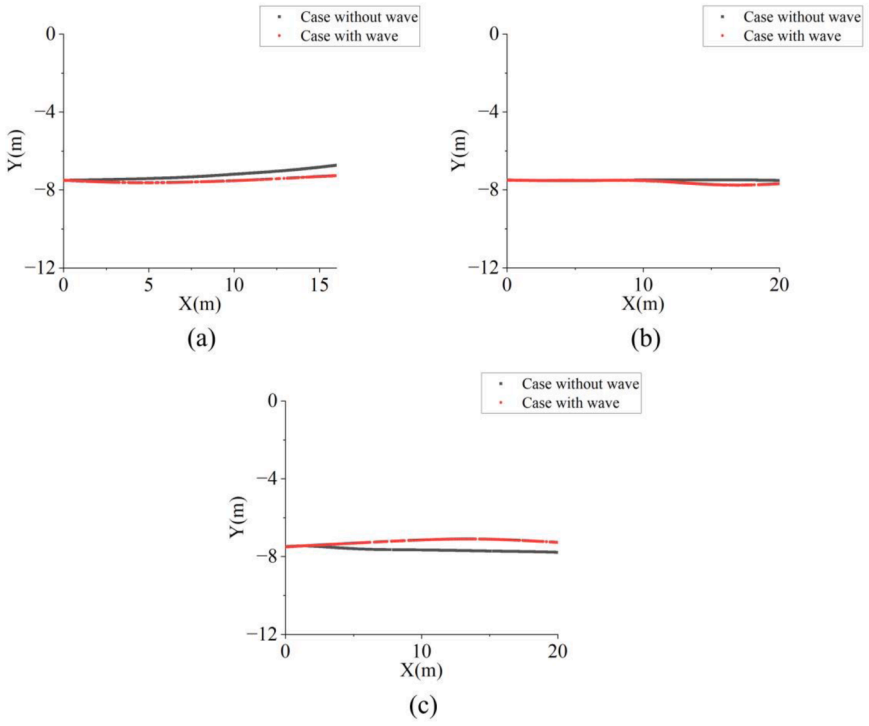

Fig. 19. The wave effects on drift trajectory of the life raft: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

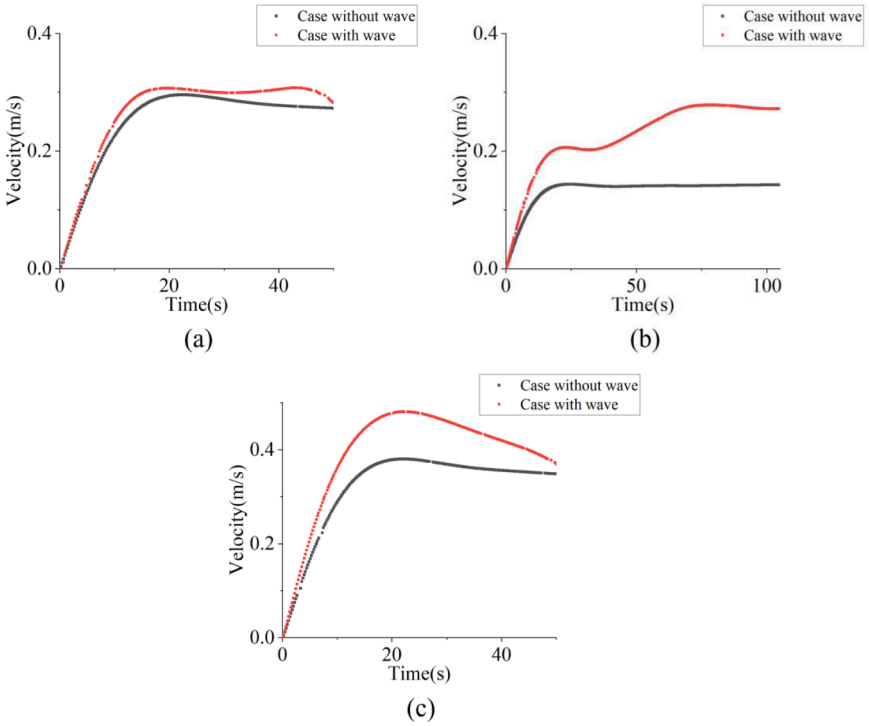

Fig. 20. The wave effects on drift velocity of the life raft in the x-direction: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

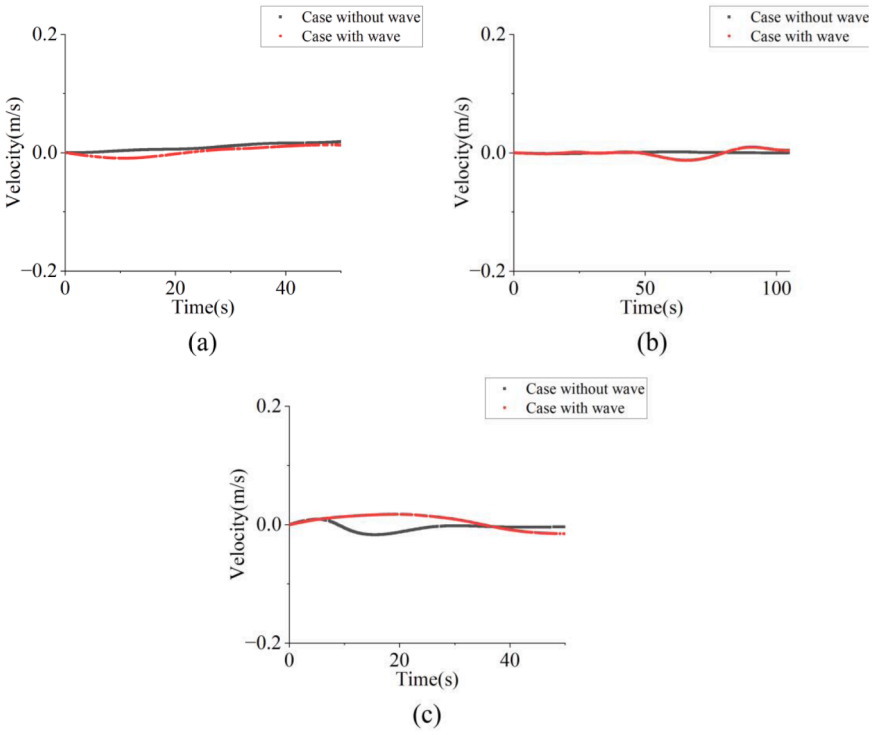

Fig. 21. The wave effects on drift velocity of the life raft in the y-direction: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

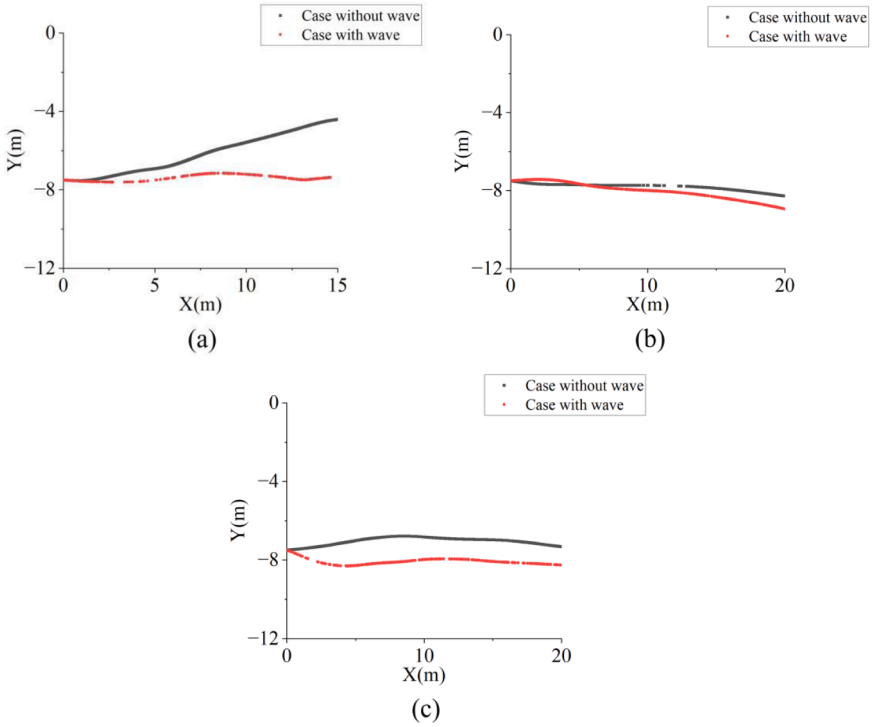

Fig. 22. The wave effects on drift trajectory of the vertical dummy: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

raft is not pronounced with respect to the drift trajectory or y-direction drift velocity. However, a comparison of the x-direction drift velocities in the wave and no-wave cases clearly indicates that waves do influence the life raft 's drift.

Figs. 22, 23, and 24 illustrate the wave effects on the drift trajectory, x-direction drift velocity, and y-direction drift velocity of the vertical dummy, respectively. Fig. 22 demonstrates that under wave influence, the drift trajectory undergoes substantial changes. These changes can be attributed to variations in the y-direction drift velocity, as depicted in Fig. 24. Furthermore, Fig. 23 confirms that the x-direction drift velocity

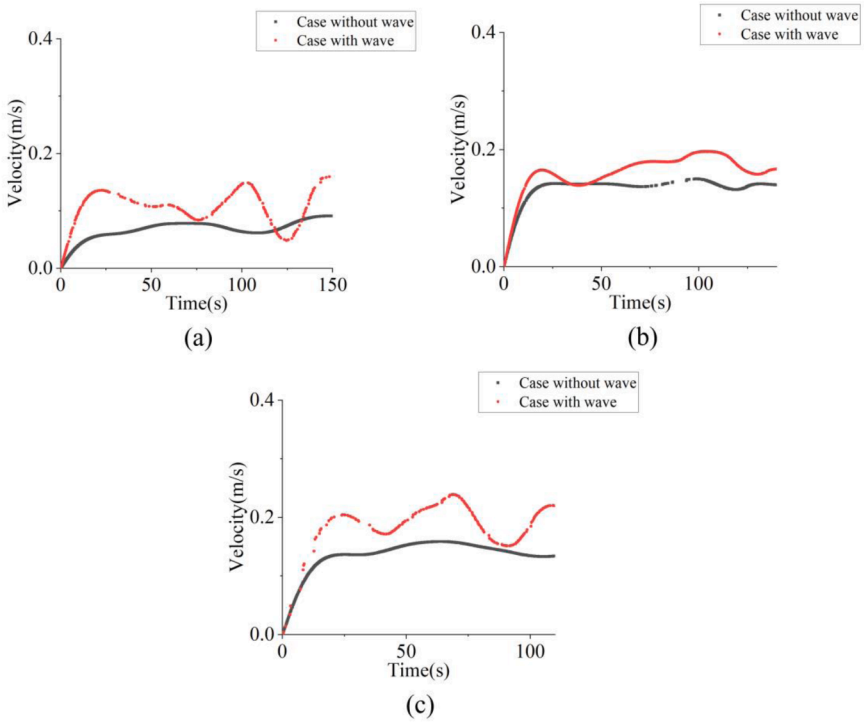

Fig. 23. The wave effects on drift velocity of the vertical dummy in the x-direction: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

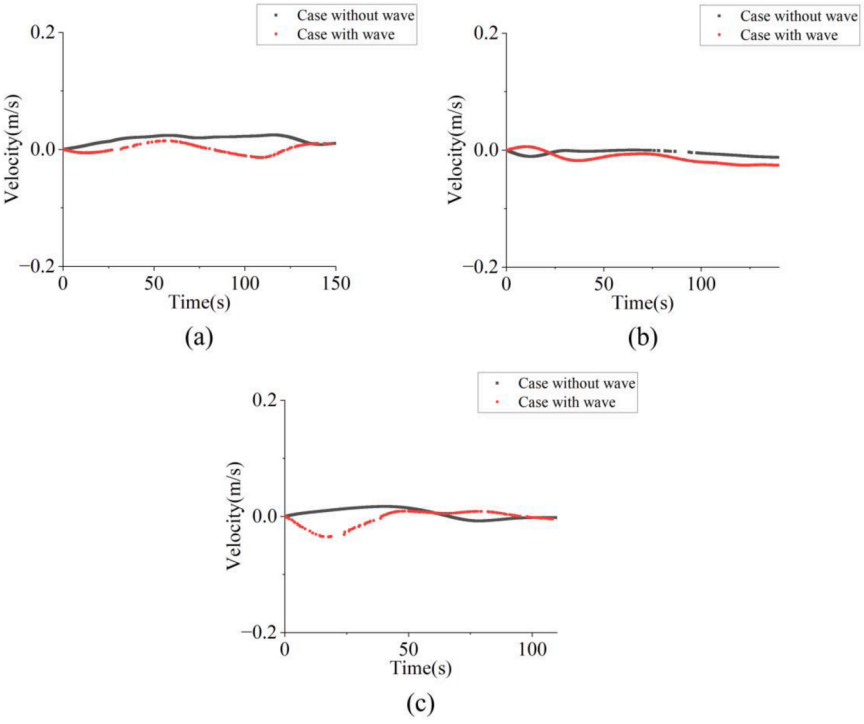

Fig. 24. The wave effects on drift velocity of the vertical dummy in the y-direction: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

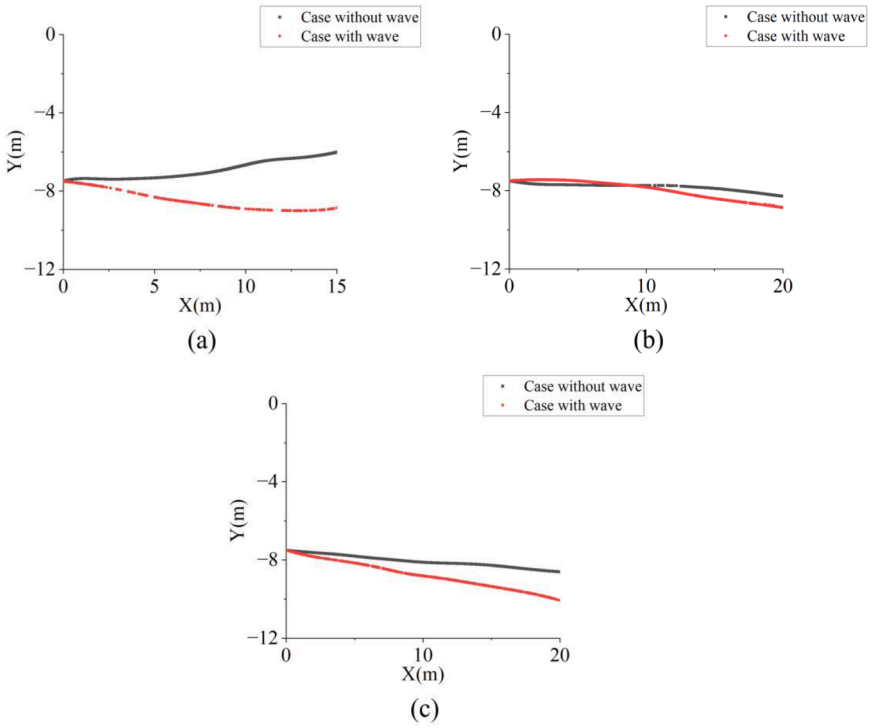

Fig. 25. The wave effects on drift trajectory of the horizontal dummy: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

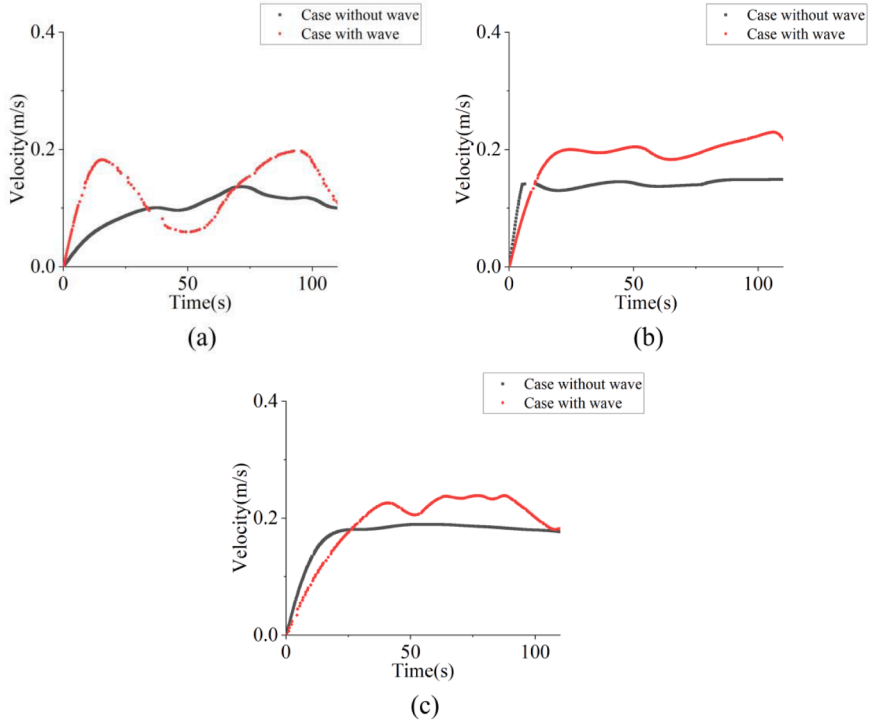

Fig. 26. The wave effects on drift velocity of the horizontal dummy in the x-direction: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

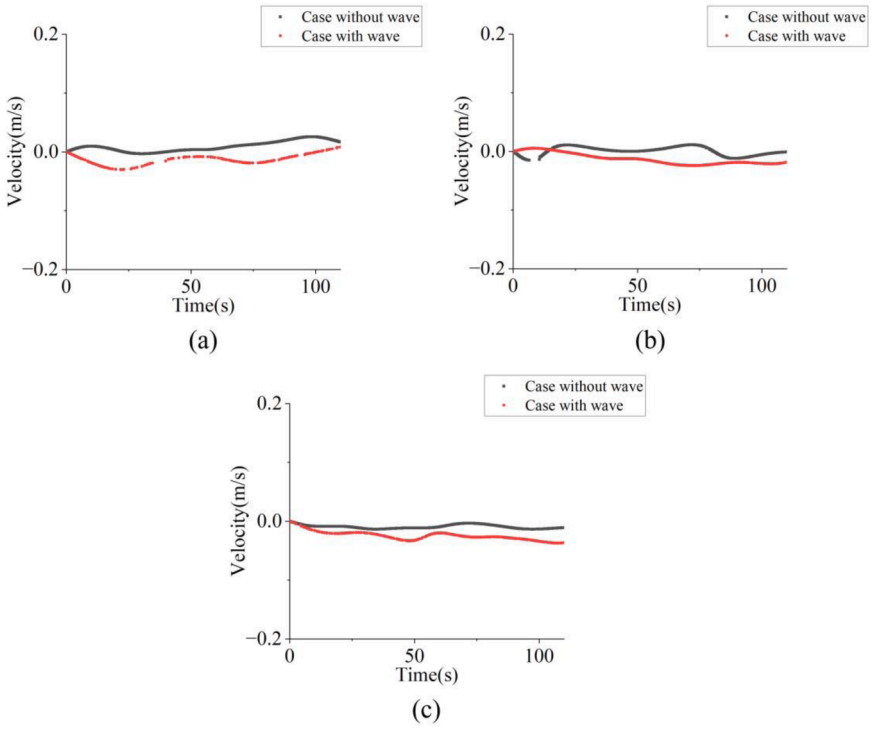

Fig. 27. The wave effects on drift velocity of the horizontal dummy in the y-direction: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

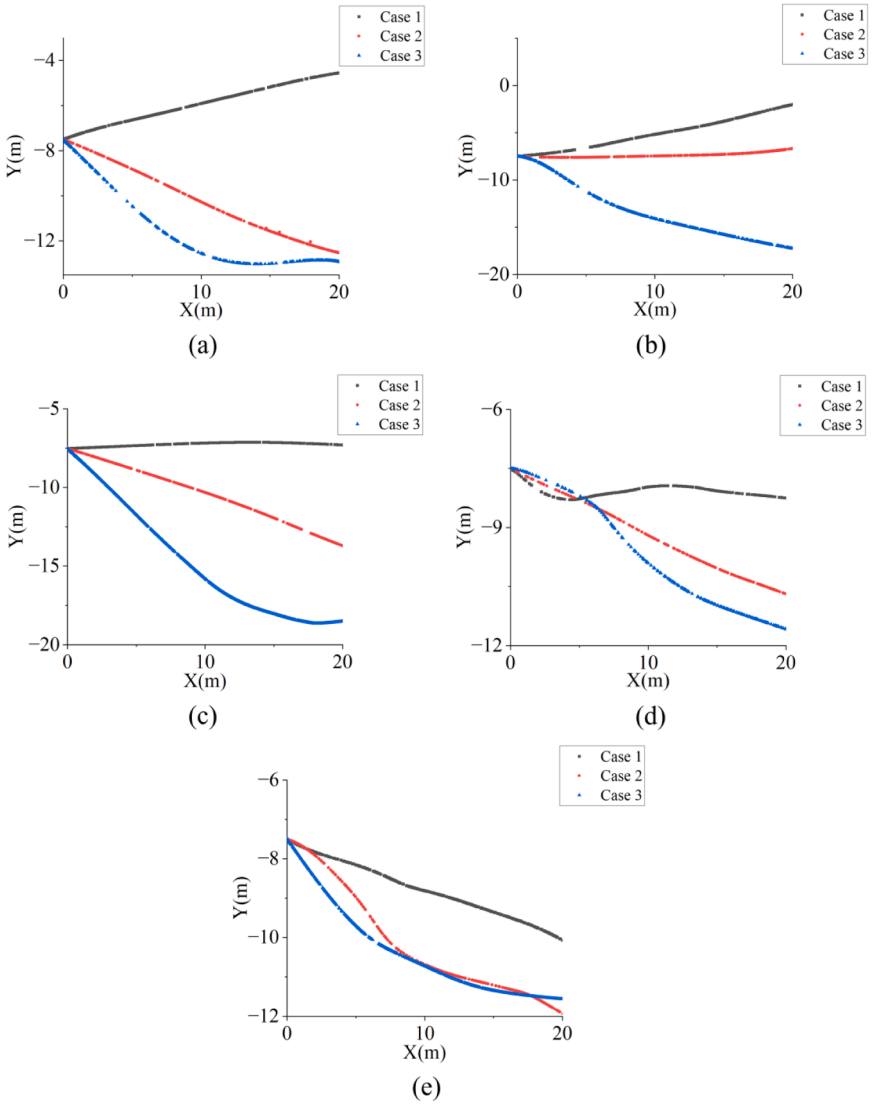

Fig. 28. Contrastive research of drift trajectories across various objects under different wind direction (Case 1: 0° wind direction; Case 2: -30° wind direction; Case 3: -60° wind direction): (a) debris; (b) fishing boat; (c) life raft; (d) vertical dummy; (e) horizontal dummy.

is greater in wave scenarios compared to no-wave scenarios, thus verifying the occurrence of wave-induced drift.

Figs. 25, 26, and 27 respectively demonstrate the wave effects on the drift trajectory, x-direction drift velocity, and y-direction drift velocity of the horizontal dummy. As shown in Fig. 25, the drift trajectory undergoes significant changes under wave action. This alteration can be attributed to variations in the y-direction drift velocity, as depicted in Fig. 27. Furthermore, Fig. 26 illustrates that the x-direction drift velocity is consistently greater in wave cases compared to no-wave cases.

3.2. Influence of wind and wave direction on object drift

3.2.1. Influence of wind direction on object drift

In this section, with the other wind, wave, and current parameters held constant, experiments are conducted using wind directions of 0°, -30°, and -60° to investigate the influence of wind direction on the drift. The specific parameter values for wind, waves, and currents are equal to those of wind, waves and current in Section 3.1. The comparison of the drift trajectories with various wind direction is illustrated in Fig. 28. As the wind direction varies, the object trajectories exhibit a significant deflection.

Fig. 29 illustrates the x-direction drift velocities with different wind direction. As the angle between the wind direction and the x-axis increases, the x-direction drift velocities of objects decrease. Once the objects drift out of the wind field, the x-direction drift velocities stabilize.

The contrastive research of y-direction drift velocities across various objects under different wind direction is studied in Fig. 30. As the angle between the wind direction and the x-axis increases, the amplitude of y-direction drift velocities increases, particularly for debris, fishing boat, and life raft. Once the objects drift out of wind field, the y-direction drift velocities tend to zero.

3.2.2. Influence of wave direction on object drift

The existing literature suggests that wind plays a significant role in influencing the deflection of object drift; however, there is a notable absence of systematic studies examining the impact of waves on drift deflection. For example, in the AP98 Leeway model, the effects of waves are integrated into the context of wind-induced drift. It is evident, though, that the drift trajectory prediction model predominantly considers the influence of wind waves on the object trajectory, while overlooking the contribution of swell. To address this gap, in this section, with the other wind, wave, and current parameters held constant, experiments are conducted using wave directions of -30°, -20°, -10°, 0°,

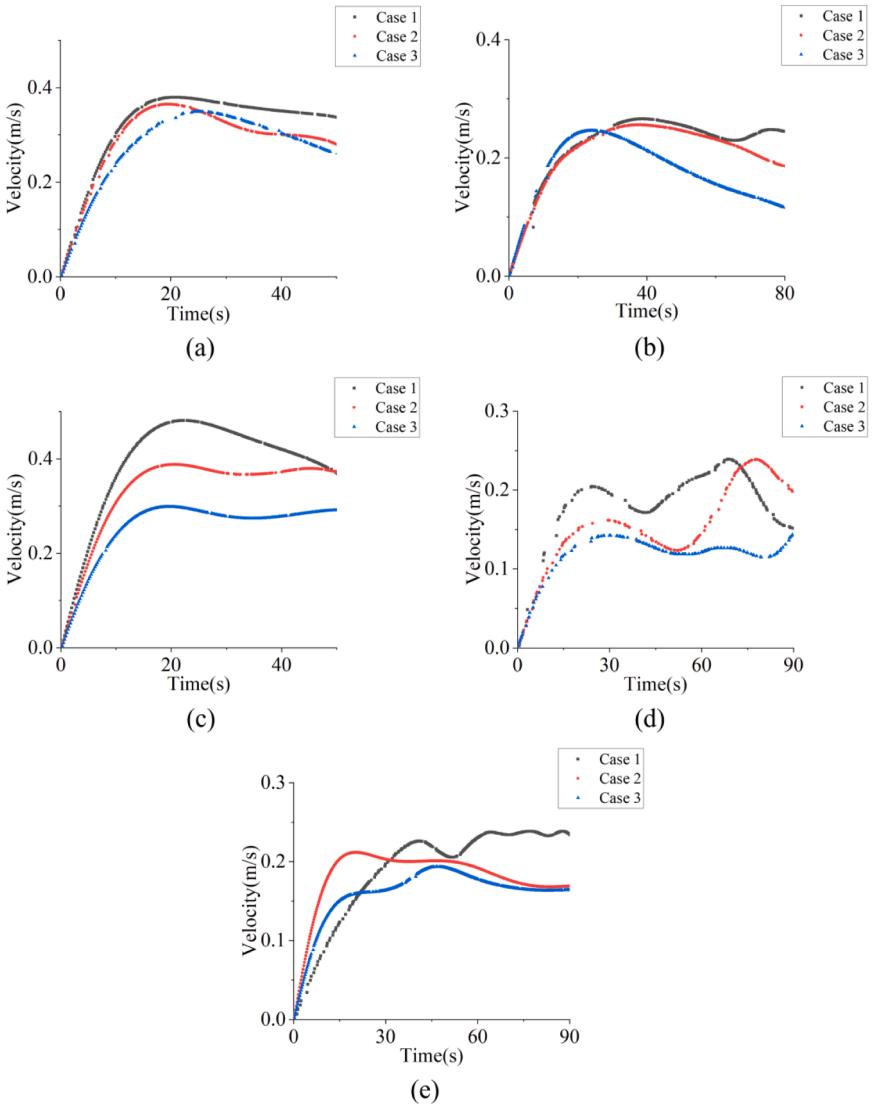

Fig. 29. Contrastive research of x-direction drift velocities across various objects under different wind direction: (a) debris; (b) fishing boat; (c) life raft; (d) vertical dummy; (e) horizontal dummy.

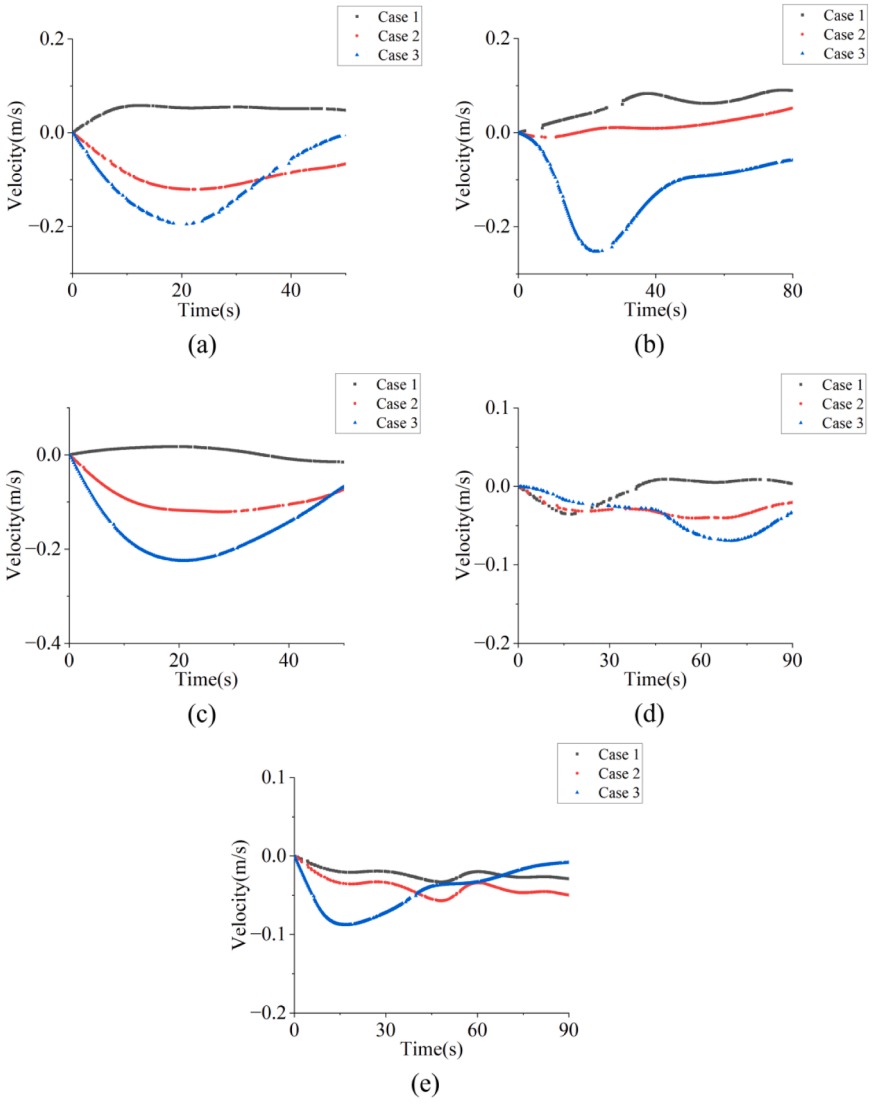

Fig. 30. Contrastive research of y-direction drift velocities across various objects under different wind direction: (a) debris; (b) fishing boat; (c) life raft; (d) vertical dummy; (e) horizontal dummy.

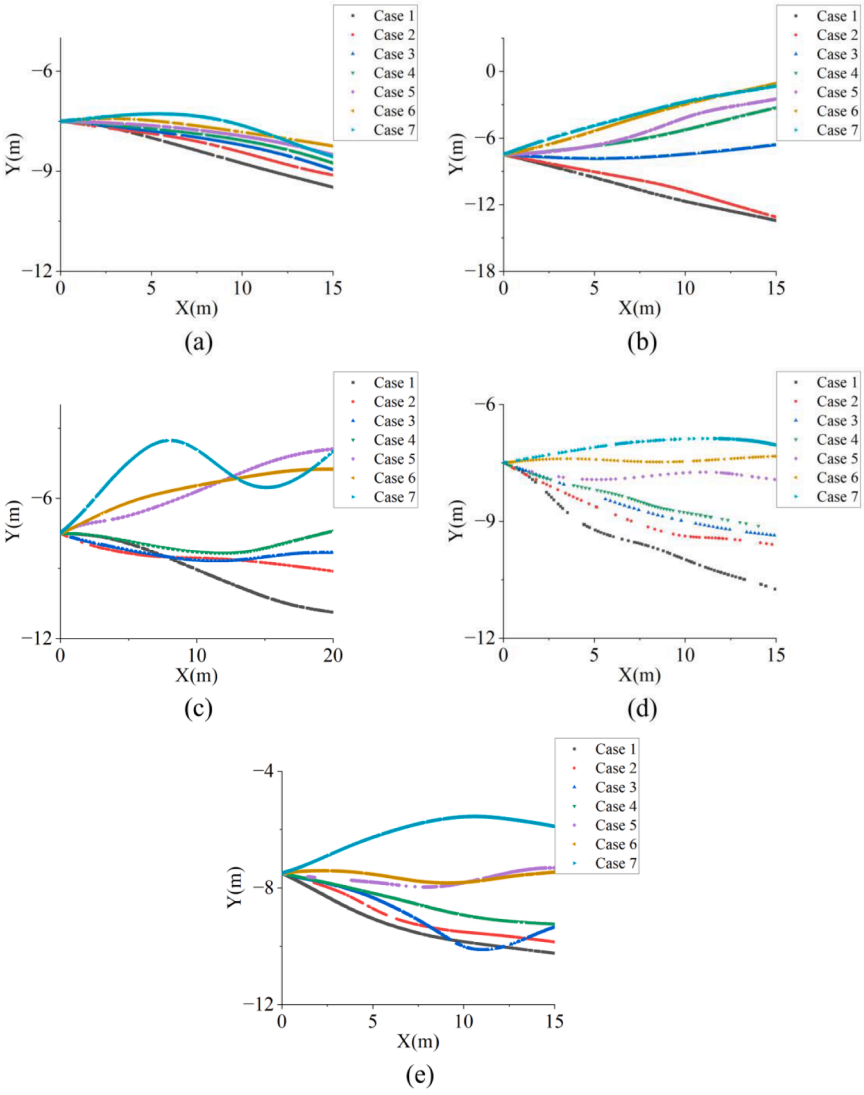

Fig. 31. Contrastive research of drift trajectories of various objects under different wave direction (Case 1: -30° wave incidence angle; Case 2: -20° wave incidence angle; Case 3: -10° wave incidence angle; Case 4: 0°wave incidence angle; Case 5: 10° wave incidence angle; Case 6: 20° wave incidence angle; Case 7: 30° wave incidence angle): (a) debris; (b) fishing boat; (c) life raft; (d) vertical dummy; (e) horizontal dummy.

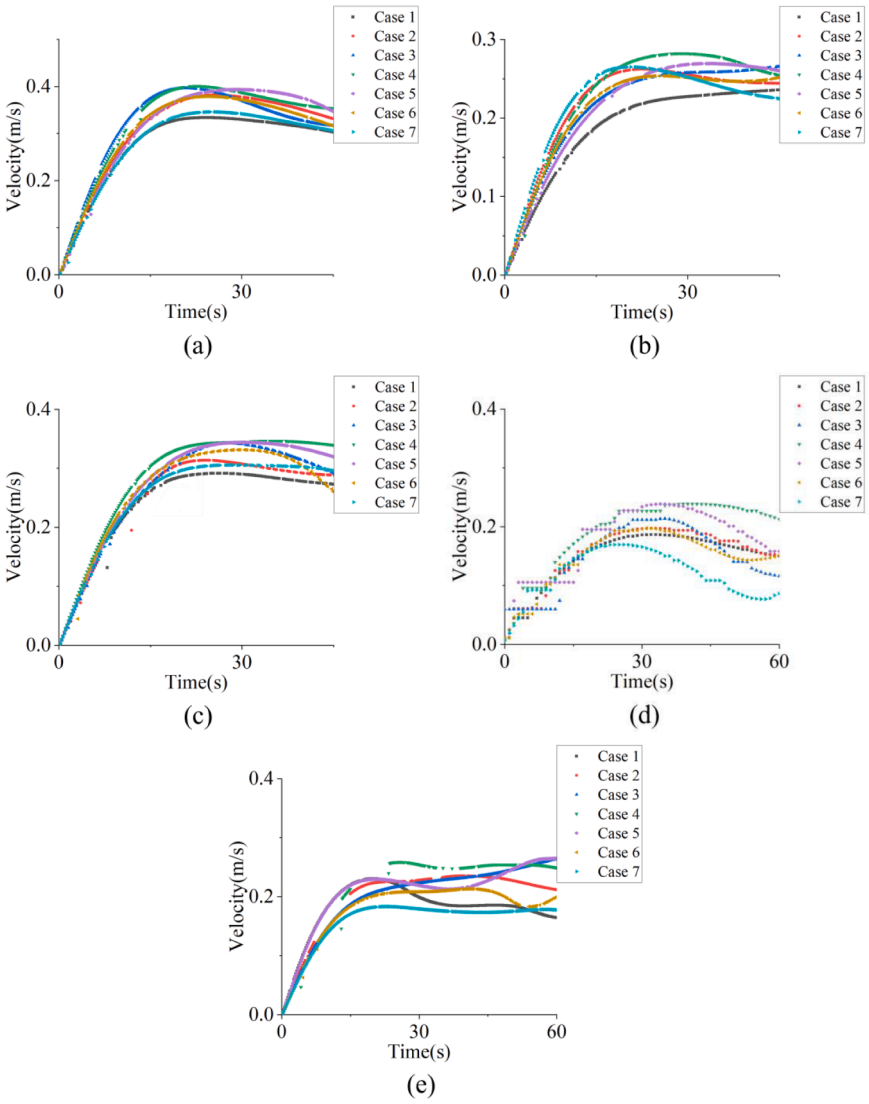

Fig. 32. Contrastive research of the x-direction drift velocities across various objects with different wave direction: (a) debris; (b) fishing boat; (c) life raft; (d) vertical dummy; (e) horizontal dummy.

10°, 20°, and 30° to investigate the influence of wave direction on the drift. In this section, the wind speed is set to 2 m/s, while the other wind, wave, and current parameters remain consistent with those in Section 3.1.

Fig. 31 depicts the contrastive research of the drift trajectories with different wave direction. It is evident that as the wave direction changes from -30° to 30°, the trajectories of the objects are deflected counterclockwise. Notably, due to the small draft of the debris, the amplitude of the deflection in its drift trajectory caused by the change in wave direction is relatively limited.

The contrastive research of the x-direction drift velocities of various objects under different wave direction is illustrated in Fig. 32. With the angle between the wave direction and the x-axis decreases, the x-direction drift velocities of objects increase. The x-direction maximum drift velocity is observed at the 0° wave direction.

The y-direction drift velocities of various objects under different wave direction are effectively depicted in Fig. 33. As the angle between the wave direction and the x-axis increases, the amplitude of y-direction drift velocities increases. The amplitude of y-direction drift velocities is the largest when the wave direction is -30° and 30°.

3.3. Drift prediction model

In the preceding chapters, we conducted the examination of the object drift trajectory. This section delves into a quantitative analysis of the drift trajectory, expanding on the established AP98 model.

Anderson et al. (1998) analyzed the force balance of marine floating objects and sorted out the theoretical framework of the Leeway model.

$$C_{iw}g\rho_w L_t A^2 \boldsymbol{C}_g \,/\, |\boldsymbol{C}_g| + C_d\rho_a A_a \boldsymbol{U}_a|\boldsymbol{U}_a| = C_{cd}\rho_w A_w \boldsymbol{U}'_0|\boldsymbol{U}'_0| \tag{3.1}$$

where: C_{iw}, C_d, and C_{cd} are the incident wave reflection coefficient, air drag coefficient for the particular object shape, and empirical water drag coefficient. g, ρ_w, and ρ_a are the gravitational acceleration, density of water, and density of air. L_t, A, A_a, and A_w are the object length scale, wave amplitude, projected frontal area of the floating object above the water's surface, and projected area of the floating object under the water. $\boldsymbol{C}_g$, $\boldsymbol{U}_a$, and $\boldsymbol{U}'_0$ are the wave group velocity, wind speed, and object velocity relative to current. $\boldsymbol{U}'_0 = \boldsymbol{U}_0 - \boldsymbol{U}_c$. $\boldsymbol{U}_0$, and $\boldsymbol{U}_c$ are the object velocity, and current speed.

Allan and Plourde (1999) proposed the AP98 Leeway model, which describes the drift velocity of a floating object at sea as the sum of its current-induced drift velocity and wind-induced drift velocity. This relationship is expressed in the following formula:

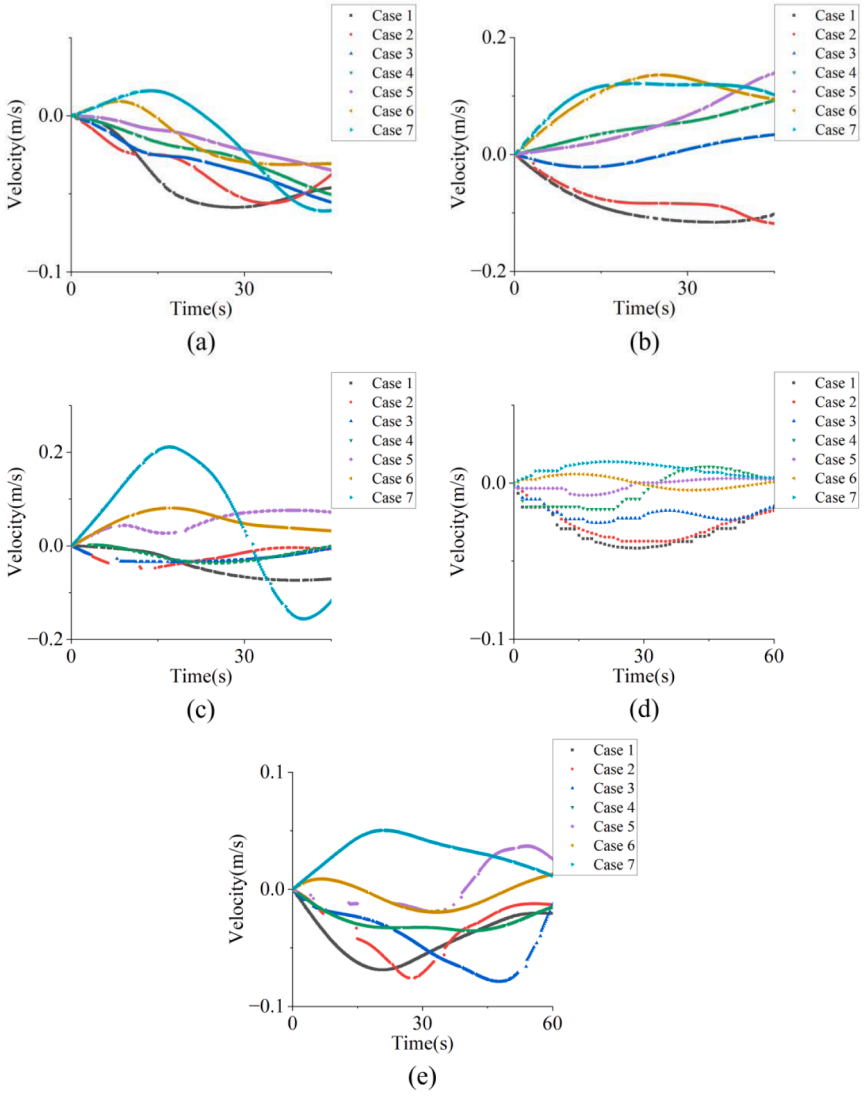

Fig. 33. Contrastive research of the y-direction drift velocities across various objects with different wave direction: (a) debris; (b) fishing boat; (c) life raft; (d) vertical dummy; (e) horizontal dummy.

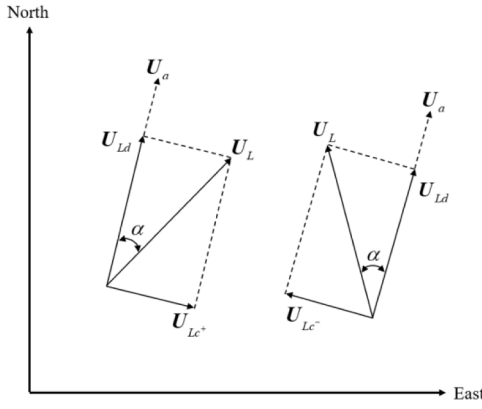

Fig. 34. Schematic diagram of the relationship between wind and wind drift.

$$\boldsymbol{U}_0 = \boldsymbol{U}_L + \boldsymbol{U}_{F-c} = \boldsymbol{U}_L + \boldsymbol{U}_c \tag{3.2}$$

where: $\boldsymbol{U}_L$, and $\boldsymbol{U}_{F-c}$ are the wind-induced drift velocity, and current-induced drift velocity. In the AP98 model, the current-induced drift velocity ($\boldsymbol{U}_{F-c}$) is regarded as equivalent to the ocean surface current speed ($\boldsymbol{U}_c$).

Allen (2005) improved the AP98 Leeway model by developing a more detailed wind drift equation, as illustrated in Fig. 34 and Eq. (3.3).

$$\begin{cases} \boldsymbol{U}_{Ld} = a_d \boldsymbol{U}_a + b_d + \varepsilon_d (DWL) \\ \boldsymbol{U}_{Lc^+} = a_{c^+} \boldsymbol{U}_a + b_{c^+} + \varepsilon_{c^+} (+CWL) \\ \boldsymbol{U}_{Lc^-} = a_{c^-} \boldsymbol{U}_a + b_{c^-} + \varepsilon_{c^-} (-CWL) \end{cases} \tag{3.3}$$

where: $\boldsymbol{U}_{Ld}$, $\boldsymbol{U}_{Lc^+}$, $\boldsymbol{U}_{Lc^-}$, and α are the downwind leeway of wind-induced drift velocity, positive (right) crosswind leeway component of wind-induced drift velocity, negative (left) crosswind leeway component of wind-induced drift velocity, and Leeway angle. a_d, a_{c^+}, and a_{c^-} are the slope. b_d, b_{c^+}, and b_{c^-} are the offset. ε_d, ε_{c^+}, and ε_{c^-} are the error term.

Six working conditions were selected to generate 1000 drift predictions for all objects, encompassing scenarios with isolated wind/current fields, wind-wave/wave-current/wind-current combinations, and wind-wave-current coupling. In this section, the wind, wave, and current parameters are held constant, and only six combinations are examined. The specific parameter values for wind, waves, and currents are equal to those of wind, waves and current in Section 3.1. The comparison and analysis of drift trajectory predictions for rescue objects under both wave and no-wave conditions using the AP98 Leeway model serve to emphasize the importance of incorporating wave environmental

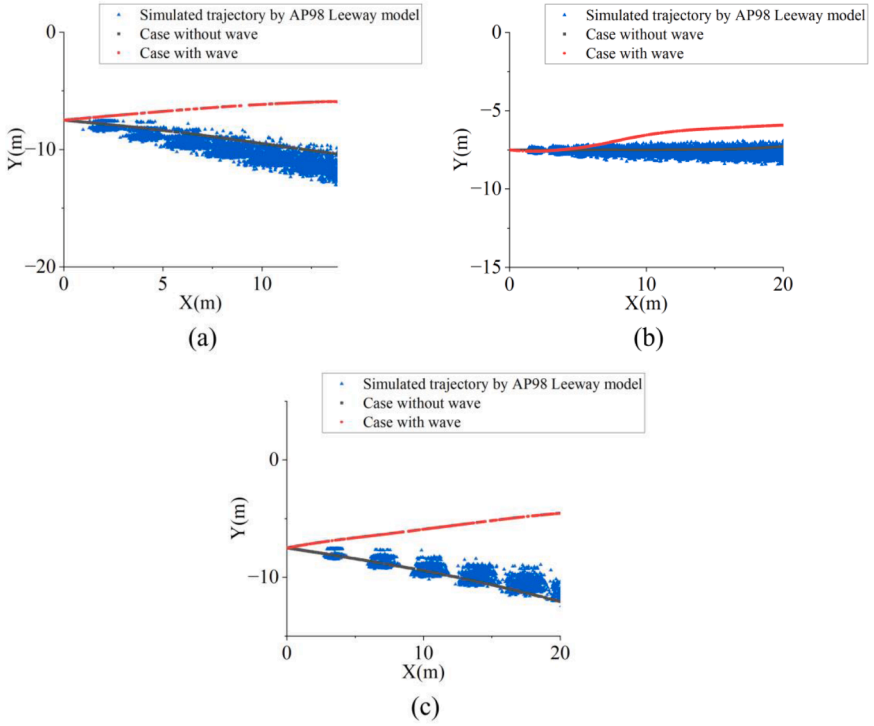

Fig. 35. Contrastive research of debris trajectories between the experimental and predicted results: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

Table 2
Average deviations of debris trajectories between the experimental and predicted results.

Case	Wind	Current	Wave	Deviation (m)
1	√	×	×	0.98
2	×	√	×	0.62
3	√	√	×	0.77
4	√	×	√	5.58
5	×	√	√	2.10
6	√	√	√	6.70

factors into drift prediction models. The assessment parameter utilized is the average deviation between the drift trajectories predicted by the AP98 Leeway model and the actual observed drift trajectories.

Fig. 35 illustrates the contrastive research of debris trajectories between the experimental and predicted results. The average deviations of debris trajectories between the experimental and predicted results are shown in Table 2. Overall, when wave influences the drift trajectory, the average deviation is relatively large, whereas in the absence of wave, the average deviation is small. The average deviations increased respectively from 0.98 m to 5.58 m, from 0.68 m to 2.10 m, and from 0.77 m to 6.70 m. This indicates that when utilizing the existing AP98 model to predict the drift trajectory of debris, it is crucial to account for the influence of waves.

Fig. 36 depicts the contrastive research of fishing boat trajectories between the experimental and predicted results. The Table 3 presents the average deviations of fishing boat trajectories between the experimental and predicted results. The average deviations increased respectively from 1.64 m to 6.29 m, from 0.64 m to 5.05 m, and from 0.93 m to 1.85 m, under the influence of varying conditions. It has been proved that for fishing boat s, the AP98 Leeway model fails to account for the impact of swell, leading to significant deviations when wave effects are considered.

Fig. 37 shows the contrastive research of life raft trajectories between the experimental and predicted results. The average deviations of life raft trajectories between the experimental and predicted results are presented in Table 4. Surprisingly, the average deviation does not exhibit significant changes regardless of the presence or absence of waves. This seems to imply that the current AP98 Leeway model for predicting the drift trajectory of life rafts may not require accounting for wave influence when the wave direction is 0°. However, it is crucial to note that under varying wave directions, the drift trajectory of the life raft undergoes substantial changes (as shown in Fig. 29(c)), which will lead to an increase in the average deviation. Therefore, when using the current AP98 Leeway model to predict the drift trajectory of life rafts, incorporating wave effects remains necessary.

Fig. 38 illustrates the contrastive research of vertical dummy trajectories between the experimental and predicted results. The Table 5 shows the average deviations of vertical dummy trajectories between the experimental and predicted results. Similar to the life raft, for the vertical dummy, the average deviation does not exhibit a significant change regardless of the presence or absence of waves. Nevertheless, it is undeniable that in the ocean, swell does not always align with the

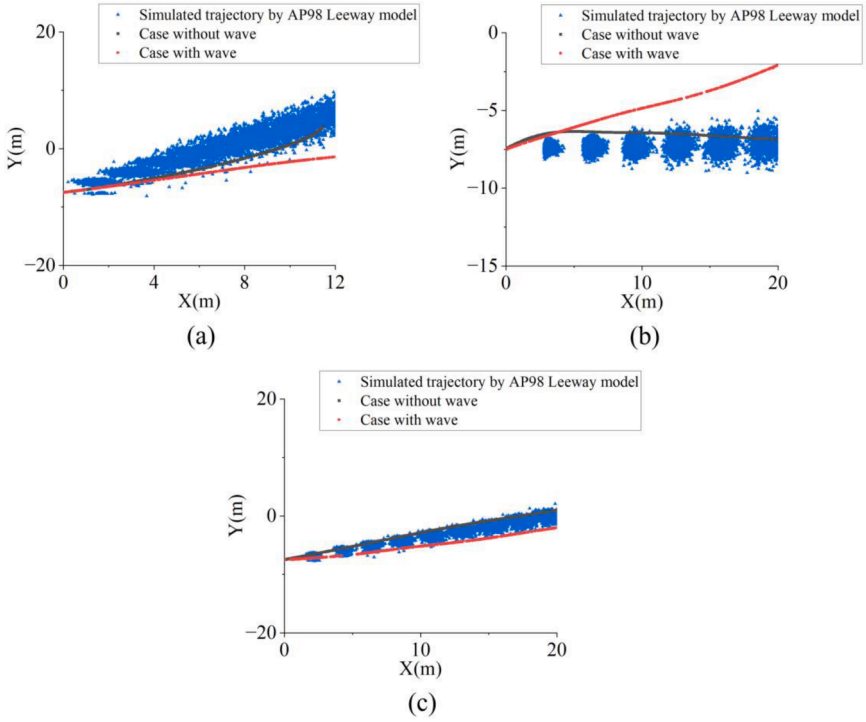

Fig. 36. Contrastive research of fishing boat trajectories between the experimental and predicted results: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

Table 3
Average deviations of fishing boat trajectories between the experimental and predicted results.

Case	Wind	Current	Wave	Deviation (m)
1	√	×	×	1.64
2	×	√	×	0.64
3	√	√	×	0.93
4	√	×	√	6.29
5	×	√	√	5.05
6	√	√	√	1.85

direction of the wind and current. Variations in wave direction will inevitably lead to an increase in the average deviation. Therefore, when using the existing AP98 Leeway model to predict the drift trajectory of the vertical dummy, it is also necessary to consider the wave effects.

Fig. 39 presents the contrastive research of horizontal dummy trajectories between the experimental and predicted results. The average deviations of horizontal dummy trajectories between the experimental and predicted results are shown in Table 6. The average deviations increased respectively from 1.32 m to 1.52 m, from 0.64 m to 1.24 m, and from 0.83 m to 2.24 m. This indicates that the existing AP98 model needs to take into account the influence of waves when predicting the drift trajectory of the horizontal dummy.

The demonstrably increased prediction error observed in the established AP98 Leeway model when applied to drift trajectories under active wave conditions stems fundamentally from its failure to incorporate wave-induced drift forces. As explicitly detailed in Eq. (3.2), the model's current mathematical formulation computes an object's total drift velocity ($\boldsymbol{U}_0$) solely as the vector sum of two components: a wind-induced leeway component ($\boldsymbol{U}_L$) and a current-induced component ($\boldsymbol{U}_{F-c}$). This $\boldsymbol{U}_0 = \boldsymbol{U}_L + \boldsymbol{U}_{F-c}$ structure inherently excludes any direct contribution from surface waves, treating them merely as an incidental factor subsumed within these primary forces or, worse, entirely disregarded.

However, substantial experimental researches in Sections 3.1 and 3.2.2 of this study demonstrate that wave effects should be taken into account when predicting the drift trajectory of objects. Specifically, the energy and momentum transferred by waves can drive significant additional drift, with quantified contributions frequently exceeding 20 % of the total drift velocity relative to scenarios dominated solely by wind and current. This magnitude of influence is non-negligible and represents a major systematic error source. The physical mechanisms are multifaceted, encompassing Stokes drift, direct wave momentum transfer pushing objects downwind or down-wave, and complex interactions where waves modify the object's effective windage and hydrodynamic drag characteristics.

Consequently, the omission of a dedicated wave-induced drift velocity term ($\boldsymbol{U}_{F-w}$) within the AP98 framework constitutes its principal limitation in dynamic sea states. The model's core structure lacks the necessary physics to accurately simulate trajectories where wave energy is significant, leading to accumulating errors over prediction horizons. The divergence between modeled trajectories and actual drift becomes particularly pronounced during storms or sustained high sea states, precisely when accurate trajectory forecasting is most critical for operations like search and rescue or pollution response. Therefore, both rigorous theoretical analysis of fluid- object interactions and robust experimental evidence converge on a single critical path forward: enhancing the AP98 Leeway model's predictive fidelity necessitates the

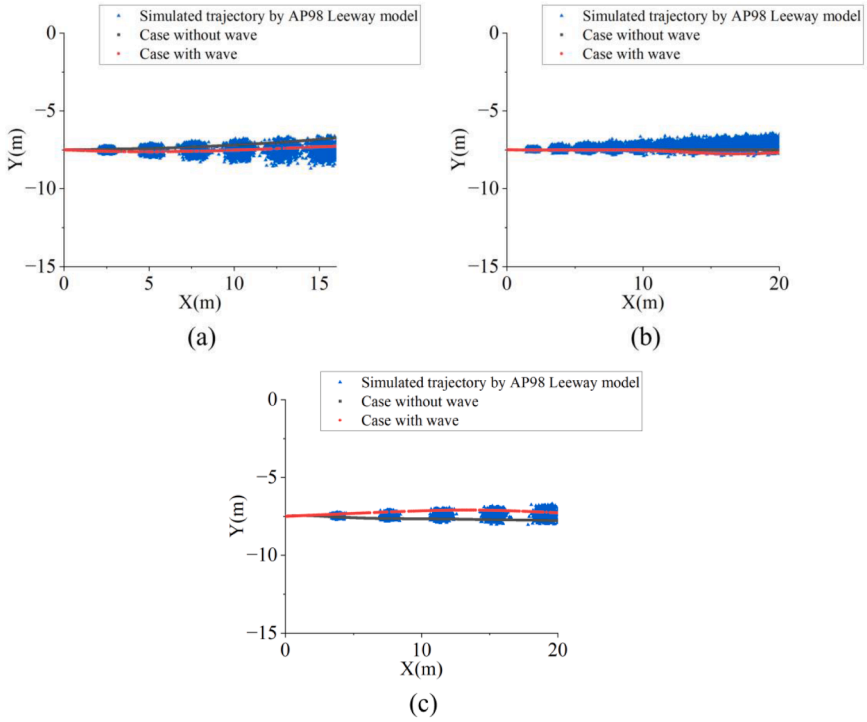

Fig. 37. Contrastive research of life raft trajectories between the experimental and predicted results: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

Table 4
Average deviations of life raft trajectories between the experimental and predicted results.

Case	Wind	Current	Wave	Deviation (m)
1	√	×	×	1.03
2	×	√	×	0.50
3	√	√	×	0.55
4	√	×	√	0.68
5	×	√	√	0.76
6	√	√	√	0.40

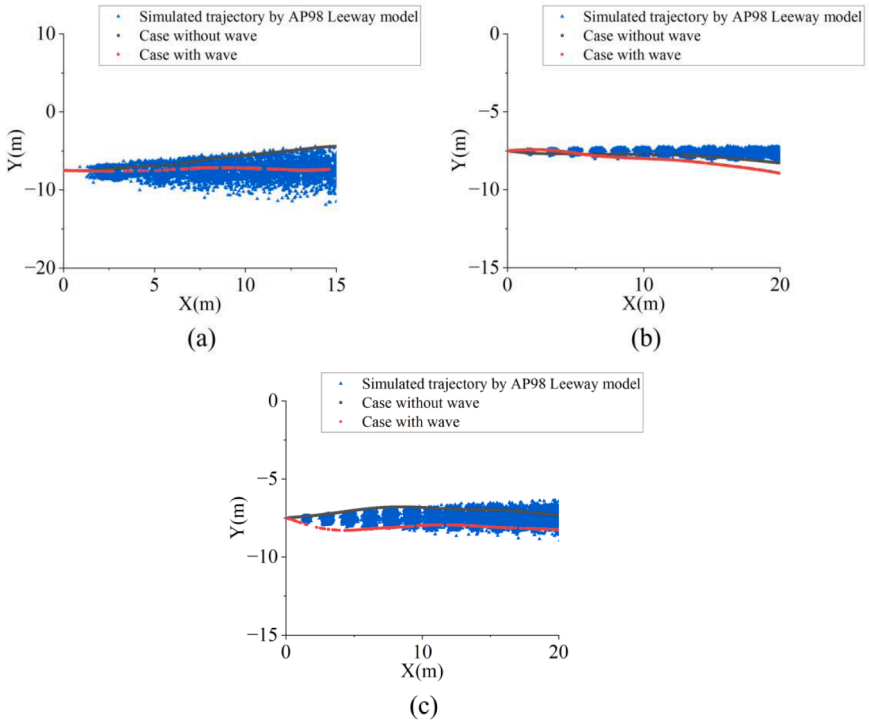

Fig. 38. Contrastive research of vertical dummy trajectories between the experimental and predicted results: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

Table 5
Average deviations of vertical dummy trajectories between the experimental and predicted results.

Case	Wind	Current	Wave	Deviation (m)
1	√	×	×	1.23
2	×	√	×	0.58
3	√	√	×	0.41
4	√	×	√	0.59
5	×	√	√	1.19
6	√	√	√	0.68

explicit integration of a wave-induced drift component. Augmenting the fundamental equation to $\boldsymbol{U}_0 = \boldsymbol{U}_L + \boldsymbol{U}_{F-c} + \boldsymbol{U}_{F-w}$ is not merely an incremental improvement but an essential step towards achieving a physically complete representation of the forces governing marine object drift.

4. Conclusions

The object drift induced by the combination of wind, wave and current is studied experimentally and numerically in this paper. The conclusions are summarized as follows:

(1) The feasibility of studying ocean drift in a small-scale experimental tank is demonstrated from the perspectives of experimental design. Compared with real sea tests, the contained hydrodynamic environment affords superior governance of exogenous variables, thereby enabling mechanistic deconstruction of drift phenomenology
(2) The study of the influence of wind direction on object drift reveals that both the deflection of the drift trajectory and the y-direction velocity of the object are significantly influenced by changes in wind direction. Additionally, the investigation into the impact of wave direction on object drift demonstrates that waves exert a substantial effect on the drift behavior of objects.
(3) The contrastive research of object trajectories between the experimental result without wave and predicted result confirms the scientific validity of the experimental methodology. Moreover, the comparison between the drift experiment with waves and the predicted trajectory indicates that accounting for wave effects is essential for improving the performance of the existing AP98 trajectory prediction model.

This investigation utilizes experimental and computational methodologies to enhance understanding of marine object drift dynamics under multi-factor environmental interactions. The integrated analysis of wind-wave-current coupling effects establishes a robust predictive framework for trajectory modeling, providing practical guidance for search-and-rescue operations and environmental monitoring. Although the current findings illustrate methodological advancements in drift forecasting accuracy, significant gaps remain in mechanistic interpretation and parameter quantification. Future research should focus on three key dimensions: 1) Implementation of high-resolution fluid-structure interaction simulations using advanced CFD techniques, 2) Expansion of test scenarios to include diverse floating object geometries and material properties, and 3) Development of multivariate environmental matrices that account for nonlinear wave interactions. Such enhancements could address existing knowledge deficiencies while improving operational prediction systems via physics-informed machine learning architectures. The proposed roadmap not only addresses current limitations in coastal hydrodynamic studies but also provides transferable methodologies applicable to marine risk assessment.

Author statement

We declare that this manuscript is original, has not been published before and is not currently being considered for publication elsewhere.

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

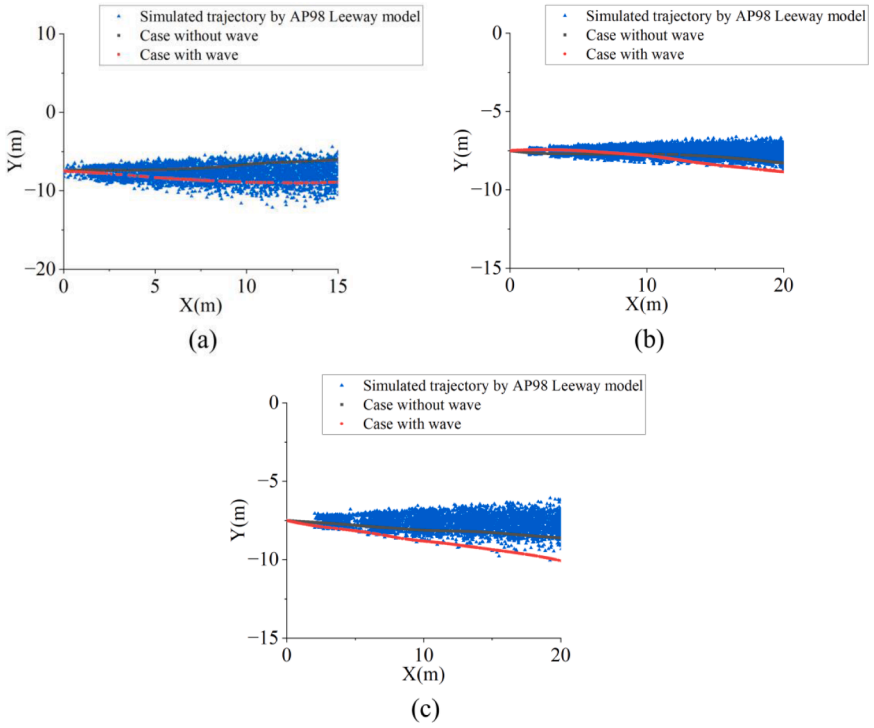

Fig. 39. Contrastive research of horizontal dummy trajectories between the experimental and predicted results: (a) isolated wind, and wind-wave; (b) isolated current, and wave-current; (c) wind-current, and wind-wave-current.

Table 6
Average deviations of horizontal dummy trajectories between the experimental and predicted results.

Case	Wind	Current	Wave	Deviation (m)
1	√	×	×	1.32
2	×	√	×	0.64
3	√	√	×	0.83
4	√	×	√	1.52
5	×	√	√	1.24
6	√	√	√	2.24

CRediT authorship contribution statement

Jiaqian Wang: Conceptualization, Formal analysis, Investigation, Methodology, Software, Writing – original draft. **Hao Qin:** Data curation, Validation, Writing – review & editing. **Daosheng Wang:** Data curation, Validation. **Zhengzhou Li:** Investigation, Resources, Software, Visualization. **Bo Li:** Investigation. **Hanxiang Su:** Investigation. **Chuanfeng Liu:** Investigation. **Lei Qin:** Investigation. **Haiwen Tu:** Formal analysis, Validation, Writing – review & editing. **Lin Mu:** Formal analysis, Funding acquisition, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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